Complex energy learning from finite magnetic-rod observations: coefficient error, local KAM and power guarantees

The International Journal of Robotics Research
XX(X):1–23
©The Author(s) 0000
Reprints and permission:
sagepub.co.uk/journalsPermissions.nav
DOI: 10.1177/ToBeAssigned
www.sagepub.com/
SAGE

Anonymous authors

Abstract

We study complex-valued energy learning for a finite-element magnetic rod with 64 transverse configuration coordinates, known constant inertia and separate potential and nonnegative dissipation heads. A deterministic observation interface converts paired short trajectories at eleven configurations into 33 force intervals and 31 derived quadratic, cubic and quartic coefficient constraints under a declared quadratic material family. Full-layer training and precision-weighted readout calibration produce a saved 918-real-parameter network. Outward evaluation verifies all constraints and bounds its actual nonzero coefficient loss below 0.010821. For the same weights and exact source operators, spectral enclosures, analyticity and a cubic-corrected twist establish local KAM for the full conservative neural Hamiltonian: every neighborhood of equilibrium contains invariant Lagrangian 64-tori whose union has positive 128-dimensional measure. Under the constitutive prior, a nonzero force-loss budget also supplies an admitted target-parameter interval with verified local KAM conditions; this is distinct from transfer to an unknown material. Dimensionally consistent complex canonical coordinates give an explicit Störmer–Verlet map with proved symplecticity and conditional fixed-time second-order convergence. Full-coordinate step-refinement calculations agree with that order. Four models matched at 914 trainable real coordinates are refitted over three seeds, providing all five real–complex and reference-energy contrasts. Separate reporting before and after readout calibration identifies substantial initialization and fitting effects. Frozen-weight predictions on six new COMSOL records at three configurations extend the comparison to a short-time numerical benchmark; the metrics do not show a consistent complex advantage. For dissipation and magnetic control, an exact power identity, potential-value allowance and sampled-work ledger give conditional energy guards. All 22 recorded numerical initial states satisfy the instantiated guard, and a native three-hold execution has a separately certified continuous reference-IVP energy cap. Physical use requires the stated constitutive, sensing, actuation and timing conditions. An exact rank calculation identifies the additional full-coordinate force information needed to complete the fourth-order jet within the four-feature target class. Independent target-tail allowances remain necessary for learning-to-KAM transfer beyond the constitutive prior.

Keywords

complex-valued neural networks, Hamiltonian learning, magnetic rods, deterministic error bounds, local KAM theory, dissipative control

1 Introduction

An energy model learned from simulated motion can support a mathematical guarantee only when the observations, learned function and downstream dynamics are defined consistently. Magnetic actuation makes this issue relevant to deformable robots: an applied field supplies mechanical work through a configuration-dependent port, while elasticity and dissipation govern the resulting motion. Magnetic manipulation and ferromagnetic soft continuum robots provide established settings for such interactions (Abbott et al. 2020; Kim et al. 2019). The question studied here is how finite observations can constrain a learned energy sufficiently to support explicitly stated conservative and dissipative conclusions.

Hamiltonian learning supplies a useful structural starting point. A conservative vector field obtained by differentiating a scalar preserves that scalar along its exact flow (Greydanus et al. 2019). This identity does not establish that the scalar approximates a specified mechanical energy. Finite trajectory observations also leave unresolved dependence away from

the observed configurations, and projection of a spatial state does not generally produce an autonomous reduced system. A learning guarantee must therefore retain observation error, the chosen mechanical chart and the distinction between a full-field model and its conservative restriction. The difficulty is especially acute when the nonlinear coefficients of interest are smaller than the uncertainty propagated from the observations.

We study a configuration potential with known kinetic energy. The mechanical object has 64 free transverse coordinates, a constant mass matrix containing translational and rotary inertia, a prescribed magnetic port, and a nominal dissipative matrix. A fixed three-dimensional trial space defines a separate conservative Hamiltonian restriction. The learned full-field correction depends on three projected coordinates and the full nominal elastic energy; on the trial space these four features depend on only three independent configuration variables. Two complex networks represent the potential correction and a nonnegative dissipation factor, with 918 real trainable parameters in total. The kinetic energy remains fixed, so the learned correction has no momentum dependence.

The observation interface retains the full spatial records. Eleven prepared configurations each have two signed-velocity records, giving 22 selected records. Their time samples, configurations, velocities and prescribed fields produce conservative-force labels and signed weak residuals. Continuous enclosures receive each force label against the declared equations initialized by its own recorded numerical state. The bounds retain reconstruction defects, preparation error and arithmetic error. They concern a specified semidiscrete mechanical system; they do not identify a numerical reconstruction with a compiled simulator equation or turn simulated observations into physical measurements.

An exact finite design then connects the 33 force components to the 31 ordinary potential coefficients of degrees two through four. Its error bound propagates componentwise force allowances through an exact left inverse and retains an independent fifth-derivative remainder for a general target. The instantiated calculation uses an explicitly declared quadratic material-correction family, for which this remainder vanishes. This family assumption is separate from the observed labels. The coefficient objective uses the transformed label centers and their received scales, allowing its actual nonzero loss to enter a deterministic candidate-to-target coefficient bound. This finite-data route addresses the nonzero-error question raised by Chen et al. (2022) without assigning an independent sampling law to correlated trajectory data.

The same constant-mass energy also admits a directly constructed symplectic difference map. Action, energy and time scales first give dimensionally consistent complex canonical coordinates. Splitting the kinetic and potential flows then yields Störmer–Verlet, with an explicit proof of symplecticity and fixed-time second-order convergence. A separate force-error estimate distinguishes learning error from time-discretization and implementation error. A fixed full-coordinate step-refinement experiment checks this construction on the saved network against an independently implemented numerical reference.

The saved candidate is obtained through full-layer training followed by precision-weighted calibration of its final energy readout. With the trained hidden layers fixed, the energy skip and the final complex readout form a 25-dimensional linear fitting block. One numerical least-squares solve calibrates this block within the unchanged architecture. Independent outward evaluation of the saved graph verifies all 33 force and all 31 coefficient interval constraints and supplies a strict upper bound 1.082009×10^{-2} for the nonzero coefficient loss. A separate source-bound calculation verifies the unchanged neural energy on all 64 transverse coordinates: a complete positive spectrum, fourth-order finite nonresonance, residual-controlled modal projections, a nonsingular cubic-corrected twist and an analytic neighborhood imply local Lagrangian 64-tori. The earlier three-coordinate restriction remains a distinct result. Under the explicit hypothesis $\mathcal{W}_* = \beta E_0$, a direct interval calculation also verifies the local criterion for every target Hamiltonian in the admitted scalar interval. A further force-loss interface uses the same saved complex graph: its nonzero weighted force error implies a source-prior target-parameter interval on which all 64 transverse conservative KAM conditions remain valid. This deterministic certificate is more conservative than direct data intersection and makes no population-risk or generic constitutive identification claim. It uses the family prior rather than a perturbative comparison with the learned candidate. For the full-field system, algebraic power identities and a uniform potential-value allowance give a conditional continuous-feedback guard. All 22 saved numerical initial states satisfy that fixed guard. A further work bound for held magnetic inputs yields a conditional sampled-feedback theorem with a strict energy ledger; all 22 nonzero initial proposals are accepted at both 1 and 10 μs hold durations. A further native execution applies two nonzero feedback holds and a scheduled zero hold. A continuous reference-IVP certificate combines residual-enclosed port errors for the two nonzero holds with the

dissipative zero-input branch, while preserving the fixed prospective error allowance and recorded commands.

The contribution is a connection between finite observations, an actual nonzero learning objective and conclusions for one specified saved model, with their different hypotheses kept explicit. Its limits are also informative. Exact interval intersection shows that the observations admit a scalar stiffness correction, while their quartic uncertainty exceeds the weak prescribed magnetic quartic. The data therefore do not require a nonlinear correction or establish an advantage of complex networks. A further fixed-budget comparison refits real and complex models, with and without a fixed quadratic output contribution, over three seeds and all five paired contrasts. The total real parameter budgets are matched after removing exact output-bias gauges. Separating the joint-training and final-readout stages reveals the roles of residual initialization and readout calibration, with different real–complex orderings across stages and error metrics. After the twelve weights are frozen, three new signed-pair COMSOL configurations test full-coordinate forced and damped prediction over $20\,\mu$s. The resulting benchmark errors remain close across the four cells. The candidate certificate covers all 64 transverse coordinates of the conservative neural Hamiltonian defined by the saved weights and exact source operators. It requires no invariant trial subspace. Runtime floating-point buffers approximate these operators; they are not identified with exact machine trajectories. A separate calculation establishes local KAM conditions for the declared quadratic target family. Neither result identifies an unknown physical constitutive law. The continuous and sampled safety theorems assume the stated physical family and full-state port. The sampled result also requires valid observations at the actual hold starts, bounded physical damping, and execution of the accepted total field with the prescribed timing. The executed simulation and its reference-IVP certificate supply no physical sensor guarantee or equivalence theorem for the compiled simulator equations. Identification of unknown weak nonlinearities, perturbative learned-to-plant transfer and physical feedback implementation remain separate requirements.

1.1 Related work

Hamiltonian and port-Hamiltonian learning. Greydanus et al. (2019) parameterize a scalar Hamiltonian and differentiate it to construct a conservative vector field. This separates conservation of the learned scalar from unconstrained vector-field regression. Zhong et al. (2020) extend structured learning to dissipation and external inputs through a port-Hamiltonian computation graph, using observed trajectories and differentiable integration. Desai et al. (2021) likewise use port-Hamiltonian structure to learn systems with time-dependent forcing and dissipation. These works motivate retaining energy, loss and supply as distinct components of a model. We fix the native kinetic and input operators, learn a configuration correction and a scalar dissipation factor, and verify the resulting saved functions independently of their optimizer. The exact power identity is an architectural property; accuracy relative to the specified target enters through separate observation and potential-error bounds.

Complex network representations. Trabelsi et al. (2018) develop practical components for deep complex networks, including complex convolutions, normalization and initialization, and evaluate them on vision and audio tasks. Their results motivate complex affine representations but do not supply a mechanical identification theorem for the present network. Zhang et al. (2022) establish approximation and optimization results for specified complex-reaction networks, including a representation separation for a class of radial functions. Those statements depend on their network and activation assumptions. Our split-tanh implementation is interpreted as a real differentiable graph before continuation in complex configuration variables. Its analytic domain and low-order coefficients are checked for the actual saved parameters. We use neither of these representation references as a generalization bound or a proof of superiority for this architecture. The scalar comparison admitted by the present observations makes that distinction consequential.

Symplectic approximation. For separable Hamiltonians, Störmer–Verlet composes exact kinetic and potential subflows into a symmetric second-order symplectic map (Hairer 2010, Lecture 2, Theorem 2 and Section 5, Example 3). The current learned correction depends only on configuration and retains constant mass, so this classical construction applies directly. The model-specific steps are the full energy-feature derivative and the physical-to-complex canonical scaling. Local truncation, learning and numerical implementation errors enter different bounds.

Nonzero learning error and conservative invariant tori. Chen et al. (2022) connect nonzero Hamiltonian learning error to KAM theory through gradient-class complexity, statistical generalization and an energy sup-norm estimate. Applying such a route requires its sampling, class and regularity hypotheses. Here the time samples, signed pairs and derived coefficients reuse a fixed finite observation design; they are not independent configuration samples. We instead propagate explicit force allowances to a

finite coefficient error and then evaluate the candidate's conservative conditions separately. Classical KAM theory requires arithmetic and nondegeneracy hypotheses in addition to a perturbation estimate (Pöschel 2001). A posteriori formulations can quantify existence near an approximate torus (Haro and Luque 2019). The saved weights and exact source operators define a full 64-coordinate neural candidate treated with a local elliptic criterion using the cubic-corrected quartic normal form. The proof bounds modal residuals before enclosing the actual twist, and retains the generic fifth-order remainder. A separate source-conditioned calculation covers all 64 transverse coordinates of the quadratic target family. The actual nonzero force loss also supplies an outer parameter set on which this direct family criterion is checked. These are qualitative local results, with no specified finite torus domain; the target-family conclusion uses its constitutive prior rather than perturbative transfer of the candidate tori to an unknown material.

Magnetic robotics and the scope of mechanical verification. The survey of Abbott et al. (2020) describes magnetic forces, torques and field generation for remote robotic manipulation. Kim et al. (2019) demonstrate remotely steerable ferromagnetic soft continuum robots. These works provide the robotic context for a mechanical input port whose work must be accounted for alongside elastic storage and dissipation. The present study addresses a declared semidiscrete beam model and simulated observations. Its full-field power and conditional safety arguments retain the same mass and magnetic port. The full-coordinate conservative neural theorem and its three-coordinate restriction answer separate dynamical questions. A verified mathematical candidate, a finite-element observation record and a physical robot remain distinct objects unless their transfer errors and implementation conditions are separately bounded.

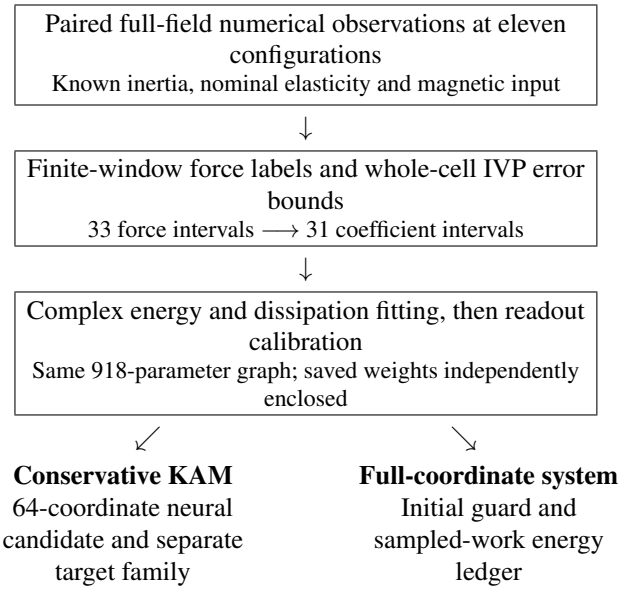


Figure 1. The observation-to-proof chain. With forcing and dissipation removed, the saved weights and exact source operators define a 64-coordinate neural candidate with a source-bound local KAM certificate. Its three-coordinate restriction is verified separately. Under the source-conditioned prior $\mathcal{W}_* = \beta E_0$, a separate proof covers the entire 64-coordinate transverse target family, including the larger parameter set obtained from the actual nonzero force loss. Full-coordinate safety uses a separate potential-value guard and continuous or sampled feedback under explicit physical and execution conditions. All 22 saved numerical initial states satisfy the guard; a further numerical feedback execution receives a continuous cap for a separately specified reference IVP over three holds. Neither branch identifies unknown physics, certifies a physical sensor or establishes complex-network superiority.

2 Mechanical model, complex canonical coordinates and symplectic differences

2.1 Mechanical object and complex energy representation

2.1.1 Full-coordinate finite-element mechanics The mechanical object is the transverse finite-element model of a clamped magnetic rod. Its 32 free nodes supply $x = (w_1, \theta_1, \ldots, w_{32}, \theta_{32})^T \in \mathbb{R}^{64}$, with transverse displacement and cross-section rotation at each node. The mesh has spacing 0.0025 m over a length of 0.08 m. The assembled matrices define a particular finite model; no equivalence to an arbitrary geometrically nonlinear continuum rod is assumed. The nominal elastic energy and the known inertia are

$$E_0(x) = \tfrac{1}{2} x^T K_0 x, \qquad M = M_0 + M_r \succ 0, \qquad D_0 = \alpha M_0 \succ 0. \tag{1}$$

Here M_r is the rotary-inertia contribution, $K_0 \succ 0$ and $\alpha > 0$. The damping matrix uses M_0, rather than the total mass. These mechanical equations use physical SI time and energy. The exact saved finite-element entries define the reference operators; the sum $M_0 + M_r$ is formed before any further rounding. The exact source formulas below define the mathematical graph used in the certificates. Training evaluates once-rounded binary64 operator buffers; the saved network weights and feature scales are interpreted as exact binary constants in the certified graph. No identity with every rounded runtime operation is assumed.

Let $b = e_{64}$ select the tip rotation. For a magnetic field u in tesla and fixed positive coefficients κ and μ, define

$$\begin{aligned} U_0(x) &= E_0(x) + \kappa[1 - \cos(b^T x)], \\ g(x) &= \mu \cos(b^T x) b. \end{aligned} \tag{2}$$

The learned full-state equations are

$$\dot{x} = M^{-1} p = v, \qquad \dot{p} = -\nabla U_N(x) - D_N(x) v + g(x) u. \tag{3}$$

The simulation target used for coefficient evaluation has correction $\mathcal{W}_*(x) = \beta E_0(x)$ and constant damping proportional to D_0. The saved configurations independently specify linear elasticity with geometric nonlinearity disabled. They justify this quadratic material structure for the ideal benchmark, separately from the observed force labels (Supplementary Section S2). The generator uses $E/E_{\mathrm{nom}} = 53/50$; this ratio is not inserted as a training label. The matrices remain the stated exact finite-source constants, without a compiled-runtime equivalence claim. The same constitutive law is not inferred for a general unknown physical device.

2.1.2 Restriction and its closure boundary Fix a full-column-rank preparation matrix $V \in \mathbb{R}^{64\times 3}$ and define

$$\begin{aligned} A &= V^T M V, \qquad B = A^{-1} V^T M, \\ K_3 &= V^T K_0 V, \qquad d = V^T b. \end{aligned} \tag{4}$$

Thus $BV = I_3$ and $q = Bx$ is the observed configuration feature. Restricting $x = Vq$ and pulling back the canonical form gives the three-degree-of-freedom Hamiltonian

$$H_N^r(q, p_r) = \tfrac{1}{2} p_r^T A^{-1} p_r + U_N(Vq). \tag{5}$$

The associated full momentum is $p = MVA^{-1} p_r$. This is a Hamiltonian restriction with a well-defined canonical structure. It is not a proof that $\mathrm{range}(V)$ is invariant under the full 64-coordinate dynamics. Full-field observations retain their full coordinates in the label construction and in the power analysis; they are not replaced by a projected autonomous trajectory.

2.1.3 The two complex-valued heads Both heads are fully connected $4 \to 12 \to 12 \to 1$ complex networks. For a hidden complex value z, the split activation is $\sigma(z) = \tanh(\mathrm{Re}\, z) + i \tanh(\mathrm{Im}\, z)$. Affine weights and biases are complex; the physical output energy is real. This implementation follows complex affine components and real/imaginary processing as in Trabelsi et al. (2018). Its realification, rather than a claim that a real-valued energy is holomorphic in the original complex hidden variable, defines the analytic continuation used below.

With saved positive scales Q_j and E_{net}, the four real features are

$$s(x) = \big(q_1/Q_1, q_2/Q_2, q_3/Q_3, E_0(x)/E_{\mathrm{net}}\big), \qquad q = Bx. \tag{6}$$

Let $N(s)$ be the real energy readout and $(R(s), I(s))$ the real and imaginary dissipation readouts. Set $\gamma_j = \partial_j N(0)$ for $j = 1, 2, 3$ and define

$$F_N(x) = c_E E_0(x) + E_{\mathrm{net}} \mathcal{N}_c(s(x)), \tag{7}$$

$$\mathcal{N}_c(s) = N(s) - N(0) - \sum_{j=1}^{3} \gamma_j s_j,$$

$$\begin{aligned} a_N(x) = {} & [1 + c_C + R(s(x)) - R(0)]^2 \\ & + [I(s(x)) - I(0)]^2, \end{aligned} \tag{8}$$

$$U_N(x) = U_0(x) + F_N(x), \qquad D_N(x) = a_N(x) D_0. \tag{9}$$

The two branches and the real scalars c_E, c_C contain 918 real parameters. Equation (7) enforces $F_N(0) = 0$ and $\nabla F_N(0) = 0$. Only the first three feature derivatives are subtracted. The linear response through the energy feature is retained and contributes to the physical quadratic term. Equation (8) enforces $D_N \succeq 0$ for every parameter vector but does not alone guarantee a positive lower bound.

The configuration feature and the full elastic-energy feature must both be differentiated. Writing $F(q, E)$ for the right-hand side of (7) gives

$$\begin{aligned} \nabla_x F_N &= B^T F_q + F_E K_0 x, \\ V^T \nabla_x F_N &= F_q + F_E V^T K_0 x. \end{aligned} \tag{10}$$

On $x = Vq$, $E_0(Vq) = q^T K_3 q/2$; on a general full trajectory the elastic-energy feature is not determined by Bx alone. This is why the same full observation enters the dissipative weak residual.

2.2 Dimensionally consistent complex canonical coordinates

The conservative subsystem of (3) has

$$\begin{aligned} H_N(x, p) &= \tfrac{1}{2} p^T M^{-1} p + U_N(x), \\ \dot{x} &= M^{-1} p, \qquad \dot{p} = -\nabla U_N(x). \end{aligned} \tag{11}$$

The potential includes the magnetic storage in (2) and the unchanged complex energy head. In particular, both terms in (10) enter its force. The 64 configuration entries include displacements and rotations; physical configurations

and momenta therefore require scaling before they can be combined into a complex canonical variable.

Choose an invertible configuration scale L_s, an action scale $\mathcal{A} > 0$ and an energy scale $\mathcal{E} > 0$. Set

$$\begin{aligned} x &= L_s Q, \qquad p = \mathcal{A} L_s^{-T} P, \\ t &= (\mathcal{A}/\mathcal{E})\tau, \\ h(Q,P) &= H_N(L_s Q, \mathcal{A} L_s^{-T} P)/\mathcal{E}. \end{aligned} \tag{12}$$

The pullback of $p^T dx$ is $\mathcal{A} P^T dQ$. After the stated time and energy normalization, the chain rule gives

$$\begin{aligned} h &= \tfrac{1}{2} P^T A_s P + \mathcal{U}(Q), \\ Q' &= A_s P, \qquad P' = -g_s(Q), \\ A_s &= \frac{\mathcal{A}^2}{\mathcal{E}} L_s^{-1} M^{-1} L_s^{-T}, \\ \mathcal{U}(Q) &= \frac{U_N(L_s Q)}{\mathcal{E}}, \qquad g_s = \nabla_Q \mathcal{U}. \end{aligned} \tag{13}$$

A prime denotes $d/d\tau$. Here $A_s \succ 0$ is a normalized inverse mass, distinct from the restricted mass A in (4).

Define the complex coordinates and the inverse transformation by

$$\begin{aligned} z &= (Q + iP)/\sqrt{2}, \quad \bar{z} = (Q - iP)/\sqrt{2}, \\ Q &= (z + \bar{z})/\sqrt{2}, \qquad P = (z - \bar{z})/(i\sqrt{2}). \end{aligned} \tag{14}$$

For $\mathfrak{h}(z, \bar{z}) = h(Q, P)$, direct differentiation yields

$$\partial_z \mathfrak{h} = (h_Q - i h_P)/\sqrt{2}, \qquad \partial_{\bar{z}} \mathfrak{h} = (h_Q + i h_P)/\sqrt{2}. \tag{15}$$

The Hamilton equations and canonical form consequently become

$$\begin{aligned} z' &= -i\partial_{\bar{z}} \mathfrak{h}, \qquad \bar{z}' = i\partial_z \mathfrak{h}, \\ \omega_s &= \sum_j dQ_j \wedge dP_j = i \sum_j dz_j \wedge d\bar{z}_j, \\ \{z_j, \bar{z}_k\} &= -i\delta_{jk}. \end{aligned} \tag{16}$$

The physical two-form pulls back to $\mathcal{A}\omega_s$. Thus the configuration–momentum transformation alone is conformally symplectic with a constant factor; the time and energy scalings in (12) give the normalized Hamilton equations. This represents a real scalar Hamiltonian in complex coordinates. The complex network hidden variables need not be canonical states.

For analytic continuation, each complex affine layer is first realified, and its split activation acts by real tanh on each component. Complexifying that real graph and the independent variables Z, W defines

$$\mathfrak{h}^{\mathbb{C}}(Z, W) = h^{\mathbb{C}}\big((Z + W)/\sqrt{2}, (Z - W)/(i\sqrt{2})\big). \tag{17}$$

The physical slice is $W = \bar{Z}$; the independent-variable extension uses no conjugation during holomorphic differentiation. This is the same realified potential used in the KAM certificate.

2.3 Derivation of the symplectic difference map

The kinetic and potential parts in (13) have the exact subflows

$$\begin{aligned} D_s(Q, P) &= (Q + sA_s P, P), \\ K_s(Q, P) &= (Q, P - s g_s(Q)). \end{aligned} \tag{18}$$

The symmetric composition $\Phi_\delta = K_{\delta/2} \circ D_\delta \circ K_{\delta/2}$ gives the classical Störmer–Verlet method (Hairer 2010, Lecture 2, Theorem 2 and Section 5, Example 3):

$$\begin{aligned} P_{n+1/2} &= P_n - \tfrac{\delta}{2} g_s(Q_n), \\ Q_{n+1} &= Q_n + \delta A_s P_{n+1/2}, \\ P_{n+1} &= P_{n+1/2} - \tfrac{\delta}{2} g_s(Q_{n+1}). \end{aligned} \tag{19}$$

Writing $\Delta t = (\mathcal{A}/\mathcal{E})\delta$ gives the physical updates $p_{n+1/2} = p_n - \Delta t \nabla U_N(x_n)/2$, $x_{n+1} = x_n + \Delta t M^{-1} p_{n+1/2}$, and $p_{n+1} = p_{n+1/2} - \Delta t \nabla U_N(x_{n+1})/2$. Both endpoint forces retain the full feature chain. Eliminating momenta yields the position difference equation

$$M(x_{n+1} - 2x_n + x_{n-1}) = -(\Delta t)^2 \nabla U_N(x_n). \tag{20}$$

In complex coordinates the individual subflows are $z \mapsto z - i s g_s(Q)/\sqrt{2}$ and $z \mapsto z + s A_s P/\sqrt{2}$, respectively, with conjugate updates. Their intermediate Q, P are recovered from (14).

Let $J = \left(\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right)$ and $G_s = Dg_s = G_s^T$. Since $A_s = A_s^T$,

$$\begin{aligned} DK_s &= \begin{pmatrix} I & 0 \\ -sG_s & I \end{pmatrix}, \\ DD_s &= \begin{pmatrix} I & sA_s \\ 0 & I \end{pmatrix}, \\ (DK_s)^T J DK_s &= (DD_s)^T J DD_s = J. \end{aligned} \tag{21}$$

Composition preserves the canonical form, including its complex representation in (16). Moreover, $K_s^{-1} = K_{-s}$ and $D_s^{-1} = D_{-s}$ imply $\Phi_{-\delta} = \Phi_\delta^{-1}$. These exact-map identities establish symplecticity and self-adjointness. The consistency and error estimate are given in Section 3.1. The underlying splitting is classical; the present adaptation uses the saved

full-coordinate energy, its complete feature derivative and dimensionally consistent canonical variables.

3 Numerical approximation and observation-based learning error

3.1 Integration error, learning error and the linear step condition

The exact shear identities concern evaluation of the mathematical map. To quantify its approximation, take a common convex configuration region containing the compared exact arcs and all kick/drift stages. Suppose there, in compatible norms,

$$\begin{aligned} &\|A_s\| \le a, \quad \|g_s\| \le G_0, \quad \|Dg_s\| \le G_1, \\ &\|D^2 g_s\| \le G_2, \qquad \|A_s P_e(\tau)\| \le V. \end{aligned} \tag{22}$$

These are domain assumptions, with $\mathcal{U} \in C^3$, rather than constants inferred from the finite training loss. For an exact arc,

$$Q_e(\delta) = Q + \delta A_s P - \int_0^\delta (\delta - s) A_s g_s(Q_e(s))\, ds. \tag{23}$$

Comparison with (19) gives $\|Q_v - Q_e(\delta)\| \le aG_1 V\delta^3/6$. For $f(s) = g_s(Q_e(s))$, differentiation gives

$$\begin{aligned} f'' &= D^2 g_s[A_s P_e, A_s P_e] - Dg_s A_s g_s, \\ \|f''\| &\le G_2 V^2 + G_1 a G_0. \end{aligned} \tag{24}$$

The trapezoidal integral error is at most $(G_2V^2 + G_1aG_0)\delta^3/12$. Replacing its exact endpoint by Q_v adds $\delta G_1\|Q_v - Q_e(\delta)\|/2$. Thus, for $0 < \delta \le \delta_0$, the phase max-norm local error is at most $C\delta^3$, where

$$C = \max\left\{\frac{aG_1V}{6}, \frac{G_2V^2 + G_1aG_0 + \delta_0 aG_1^2 V}{12}\right\}. \tag{25}$$

The shear Lipschitz bounds give $\mathrm{Lip}(\Phi_\delta) \le (1 + \delta G_1/2)^2(1 + \delta a) \le e^{\Lambda\delta}$, with $\Lambda = G_1 + a$. Iteration of the local estimate consequently yields, for $T = N\delta$,

$$\|\Phi_\delta^N(y_0) - \varphi_T^N(y_0)\|_\times \le C\delta^2 \frac{e^{\Lambda T} - 1}{\Lambda}. \tag{26}$$

This proves fixed-time second-order convergence while all compared arcs and stages remain in the specified region. Supplementary Section S1 gives the full domain, scaling and accumulated implementation-error argument.

For a target with the same kinetic energy, suppose its normalized force g_* satisfies $\sup\|g_s - g_*\| \le \varepsilon$ on a common solution region, and one complete vector field has Lipschitz bound L_*. Grönwall's inequality gives

$$\|\varphi_T^N(y_0) - \varphi_T^*(y_0)\|_\times \le \varepsilon \frac{e^{L_* T} - 1}{L_*}, \tag{27}$$

with quotient T when $L_* = 0$. Combining (26) and (27) separates the second-order integration term from the learning term. A finite-node force allowance must first be extended to the required uniform force bound before this target comparison applies.

The received full-coordinate Hessian also constrains the physical linearized step. In mass-normalized modes, the Verlet matrix has determinant one and trace $2 - (\Delta t\omega_j)^2$. Its powers are bounded precisely for $0 < \Delta t\omega_j < 2$; the endpoint $\Delta t\omega_j = 2$ has a nontrivial Jordan block. The certified upper frequency bound therefore gives the sufficient physical condition

$$0 < \Delta t < \frac{35184372088832}{2441081236299847251}\ \mathrm{s} \simeq 1.44134 \times 10^{-5}\ \mathrm{s}. \tag{28}$$

This is an equilibrium linearization condition. Nonlinear error control still uses the domain and derivative bounds above. The continuous KAM theorem in Section 4 concerns the Hamiltonian flow; a discrete-map KAM theorem additionally requires step-dependent nonresonances and its own estimates. Dissipation and magnetic work enter the separate power argument in Section 5.

3.2 Finite-time integration of the saved full model

We test the conservative kick–drift–kick map on the saved 918-real-parameter complex-network checkpoint, retaining all 64 configuration coordinates and their 64 canonical momenta. The implementation restores every saved model parameter and once-rounded buffer without alteration. Each force evaluation includes the derivative through both the projected coordinates and the elastic-energy feature. Three initial states are reused from the first, middle and final samples of the same recorded positive original-configuration experiment, at recorded times 0, 10 and 20 μs, respectively. The subsequent trajectories have zero magnetic input and no damping; they are integrations of the learned conservative model rather than comparisons against the original forced COMSOL trajectories.

The protocol fixes $T = 256\,\mu$s and step sizes $h = 4, 2, 1, 0.5\,\mu$s before any trajectory is calculated. These steps are below the separately established origin-linearization threshold; that condition alone does not prove nonlinear stability. Let K_N denote the symmetrized floating-point

Table 1. Maximum normalized full-state Verlet errors over $T = 256\,\mu$s. Cases A, B and C use recorded samples at 0, 10 and 20 μs, respectively. No state dimension is discarded during integration.

h (μs)	A	B	C
4	$2.26986\,10^{-4}$	$7.81926\,10^{-3}$	$1.20772\,10^{-2}$
2	$5.61844\,10^{-5}$	$1.92320\,10^{-3}$	$2.99755\,10^{-3}$
1	$1.40087\,10^{-5}$	$4.79350\,10^{-4}$	$7.46137\,10^{-4}$
0.5	$3.49932\,10^{-6}$	$1.19730\,10^{-4}$	$1.86332\,10^{-4}$

force Jacobian at the origin and define the positive quadratic diagnostic

$$\|(\Delta x, \Delta p)\|_N^2 = \Delta x^T K_N \Delta x + \Delta p^T M^{-1} \Delta p.$$

At 65 common times we measure the maximum error against an independently integrated reference, divided by $\|(x_0, p_0)\|_N$. The reference uses DOP853 in invertible mass-whitened full-state coordinates, with a separately realified NumPy implementation of the complex-network gradient. The initial reference settings are $(\mathrm{rtol}, \mathrm{atol}, h_{\max}) = (10^{-10}, 10^{-12}, 2\,\mu\mathrm{s})$; the reference used in Table 1 uses $(10^{-12}, 10^{-14}, 0.5\,\mu\mathrm{s})$. Their maximum normalized discrepancy over the three cases is 2.34×10^{-12}, and the largest ratio of this discrepancy to the corresponding finest-step Verlet error is 2.23×10^{-7}. This reference comparison is numerical evidence, not a certified bound on the exact-flow error.

All nine adjacent observed orders $\log_2(\varepsilon_h/\varepsilon_{h/2})$ lie between 2.001 and 2.024, consistent with the proved finite-time second-order statement. At 15 prescribed state checks, the largest mass-weighted relative force discrepancy between NumPy and the original PyTorch implementation is 3.21×10^{-16}; the corresponding largest relative potential discrepancy is 2.45×10^{-11}. Over all 12 forward–backward calculations, the largest normalized recovery error is 2.40×10^{-14}. At the finest step the maximum absolute relative Hamiltonian drifts in cases A, B and C are 3.83×10^{-10}, 2.48×10^{-7} and 8.09×10^{-7}, respectively, with floating-point evaluation error included. These diagnostics establish finite-time implementation behaviour for the selected saved-model trajectories. They do not certify a discrete KAM torus, exact energy conservation, long-time stability, agreement with unknown physical dynamics or an advantage over real-valued networks.

3.3 Finite observations and coefficient error bounds

3.3.1 Paired trajectories and finite-window force labels

The data are native COMSOL outputs at eleven prescribed configurations. Each configuration has two signed initial velocities and 257 saved full-field observations over $20\,\mu$s. The eight added configurations use $q = \ell z^{(i)}$, $\dot q = \pm 25 q\,\mathrm{s}^{-1}$ and internal step 1.953125×10^{-8} s. The preceding three retain their own prescribed velocities and finest saved steps. These choices are documented before their respective computations. The source model is unchanged, and the drive is the source-decimal function

$$B_y(t) = 10^{-4} \cos(11.41692154656479\, t)\ \mathrm{T}. \tag{29}$$

The generator's material is public in the simulation protocol. No material factor or analytic target coefficient is supplied to the learning worker, but this separation is not a hidden-target test.

The finite momentum-and-magnetic-force label approximates a conservative force at a prepared configuration. A trajectory average cannot be identified with its initial value without a temporal allowance. For signed records, let $Y(T)$ be the average obtained from projected momentum increments and magnetic-force quadrature, with the known nominal elastic force subtracted. We use the fixed three-window combination

$$\widehat{f} = \tfrac{8}{3} Y(T) - 2Y(2T) + \tfrac{1}{3} Y(4T). \tag{30}$$

For an exact smooth signal, the constant term is preserved and the first two temporal moments cancel. If its third derivative is bounded by B_3, the remaining temporal bias is at most $B_3 T^3/3$. Preparation defects and finite observation errors must also be included; they are not estimated from a small training residual. The full statement and its nonnegative Peano kernel are proved in Supplementary Section S5.1.

Continuous error is received against an explicitly declared 64-coordinate initial-value problem starting at each saved numerical first row. Its shifted elastic matrix is $K = (53/50)K_0$, and its damping is $D = \alpha_{\mathrm{shift}} M_0$, with the saved binary coefficient $\alpha_{\mathrm{shift}} \simeq 0.6389230223455499\,\mathrm{s}^{-1}$. These are the declared evaluation-IVP coefficients, not extra material labels given to the learner. A cubic Hermite curve through saved positions and velocities has a computable whole-cell equation defect. In the mechanical energy metric the trajectory-error radius obeys

$$r'(t) \le a r(t) + \rho(t), \qquad r^2 = e_x^T K e_x + e_v^T M e_v, \tag{31}$$

where ρ bounds the mass-dual equation defect. Outward Bernstein-polynomial bounds, trigonometric remainders and the exact source-decimal input enclose each whole time cell. A projected-force average retains cancellations that

a generic endpoint-momentum bound would discard. The resulting received label allowances are below 4.583×10^{-8} in normalized force at all eleven chosen configurations. This certifies the declared recorded-state IVPs; it does not certify physical measurement error or equivalence to every compiled COMSOL operation. Supplementary Sections S5.3 and S5.8 give the full construction.

3.3.2 The eleven-node fourth-order design Set $q = \ell z$, $\ell = 10^{-3}$ m, and use the exact decimal energy scale $E_s = 6 \times 10^{-12}$ J for the force/coefficient interface. The saved binary scale E_{net} is kept distinct from E_s. For the restricted correction $W_*(q) = \mathcal{W}_*(Vq)$, define $w_*(z) = W_*(\ell z)/E_s$. The normalized force labels approximate $\nabla_z w_*(z^{(i)})$ at the ideal rational nodes, while the learner's once-rounded input coordinates are recorded separately.

Under $\nabla w_*(0) = 0$ and a componentwise fifth-derivative bound M_5, its ordinary Taylor coefficients of degrees two through four form $c_* \in \mathbb{R}^{31}$. With the fixed gradient design matrix $G \in \mathbb{Q}^{33\times 31}$ and exact left inverse J, one has

$$\begin{aligned} JG &= I_{31}, \\ |c_{*,\alpha} - (J\widehat{y})_\alpha| &\le \sum_{i,j} |J_{\alpha,(i,j)}|\varepsilon_{ij}, \\ \varepsilon_{ij} &= \eta_{ij} + \frac{M_5}{24}\|z^{(i)}\|_1^4. \end{aligned} \tag{32}$$

Here $\widehat{y} \in \mathbb{R}^{33}$ contains the saved force midpoints and η includes each component's continuous and arithmetic allowance. The exact full-rank construction and tail derivation are in Supplementary Proposition S11. The fifth-derivative bound is an independent target hypothesis. For the explicitly declared family $W_* = \beta q^T K_3 q/2$, $M_5 = 0$ and the origin gradient vanishes by that constitutive assumption.

Writing $\widehat{c} = J\widehat{y}$, the largest received radii for ordinary coefficients of degrees two, three and four are

$$\begin{aligned} \max_{|\alpha|=2} r_\alpha &\le 4.282817 \times 10^{-7}, \\ \max_{|\alpha|=3} r_\alpha &\le 8.141379 \times 10^{-7}, \\ \max_{|\alpha|=4} r_\alpha &\le 1.808799 \times 10^{-6}, \end{aligned} \tag{33}$$

respectively, rounded upward. These are deterministic error intervals from the same 33 observations, not 31 additional independent samples. A small force error does not by itself bound arbitrary higher derivatives; the design, centering and remainder hypotheses in (32) are essential.

3.3.3 Nonzero-loss bounds for the saved network Let $c_{\theta,\alpha}$ be the ordinary normalized correction coefficients of the actual saved graph. The mathematical data centers are $\widehat{c} = J\widehat{y}$. Their saved training centers $\widetilde{c}$ are binary numbers. Exported radii satisfy $\widetilde{r}_\alpha \ge r_\alpha + |\widetilde{c}_\alpha - \widehat{c}_\alpha|$, including the exact center-rounding allowance. With the fixed saved scales $\widetilde{s}_\alpha = \max(\widetilde{r}_\alpha, 10^{-8})$, the actual coefficient objective is

$$\widetilde{L}_{\mathrm{jet}}(\theta) = \frac{1}{31}\sum_\alpha \left(\frac{c_{\theta,\alpha} - \widetilde{c}_\alpha}{\widetilde{s}_\alpha}\right)^2. \tag{34}$$

The data interval gives, for every parameter vector,

$$|c_{\theta,\alpha} - c_{*,\alpha}| \le |c_{\theta,\alpha} - \widetilde{c}_\alpha| + \widetilde{r}_\alpha \le \widetilde{s}_\alpha\sqrt{31\widetilde{L}_{\mathrm{jet}}(\theta)} + \widetilde{r}_\alpha. \tag{35}$$

No optimizer convergence or independent-sampling assumption is used. Each component's direct residual normally gives the tighter first bound. The exact original-center version is also valid with $\widehat{c}$, r and any fixed positive scales, but is not silently substituted for the saved-center objective actually optimized.

Float64 losses guide optimization but are not proof bounds. For an outward coefficient enclosure $[a_\alpha, b_\alpha]$, define $\Delta_\alpha = \max(|a_\alpha - \widehat{c}_\alpha|, |b_\alpha - \widehat{c}_\alpha|)$. Then $|c_{\theta,\alpha} - c_{*,\alpha}| \le \Delta_\alpha + r_\alpha$. Replacing $\widehat{c}_\alpha$ by $\widetilde{c}_\alpha$ in the endpoint residual gives $\widetilde{\Delta}_\alpha$; then $(1/31)\sum(\widetilde{\Delta}_\alpha/\widetilde{s}_\alpha)^2$ is a rigorous upper bound for the actual saved-center loss. Supplementary Section S10 proves this finite-loss interface and distinguishes original from rounded training centers.

The configuration-potential formulation also permits a conditional population-risk route inspired by Chen et al. (2022). That route requires independent configuration samples, a justified bounded loss class, a density lower bound and target regularity. The eleven deterministic preparations do not instantiate those statistical assumptions. The present numerical conclusions use (32)–(35); the separate population estimates and their prerequisites are retained in Supplementary Section S3.

3.4 Full-coordinate information outside the prepared trial space

The preceding force and coefficient bounds concern the restriction $x = V\ell z$. To determine what information a full-coordinate extension would require, consider the additional target class $W(x) = E_s\Phi(z, e)$, with $z = Bx/\ell$ and $e = E_0(x)/E_s$. This is the network's feature geometry; factorization of an unknown material through these four features is an independent hypothesis. The binary network scales are retained through an invertible constant change of feature variables.

Give z_j weight one and e weight two. The material jet of physical degrees two through four then has 42 coefficients.

On the trial space,

$$a(z) = \frac{\ell^2}{2E_s} z^T K_3 z, \qquad \mathcal{A}(z, e) = e - a(z) = 0. \quad (36)$$

Restriction has rank 31 and an eleven-dimensional kernel consisting of $\mathcal{A}$ times all polynomials of weighted degree at most two. The current projected force labels and their derived coefficients therefore do not receive the full 42-coordinate jet.

Exact-source algebra for the actual eleven nodes gives

$$\begin{aligned} \operatorname{rank} G_{\text{projected}} &= 31, \\ \operatorname{rank} G_{\text{complete}} &= 41, \\ \ker G_{\text{complete}} &= \operatorname{span}\{\mathcal{A}^2\}. \end{aligned} \quad (37)$$

Here G_{complete} evaluates all 64 instantaneous force components at the same nodes. Nonzero minors over two fixed primes, with all rational denominators received, establish the lower ranks; explicit polynomial kernels supply the matching upper ranks. The current force-error certificate covers the three projected channels. The saved full-field records could support a new complete-force receiver, but do not already constitute certified additional labels.

If those complete forces are received, one further normal-direction label outside the trial space completes the polynomial design to rank 42. Indeed, choose $Bw = 0$, $V^T K_0 w = 0$, $w \neq 0$, and put $k_w = w^T K_0 w$. At $x = V\ell z + tw$, $t \neq 0$,

$$\frac{w^T \nabla W(x)}{t k_w} = \Phi_e\left(z, a(z) + \frac{t^2 k_w}{2E_s}\right). \quad (38)$$

At $z = 0$ this detects the remaining $\mathcal{A}^2$ coefficient. Without complete-force reuse, eleven such scalar labels at a specified quadratic interpolation design complete the existing restricted jet. Supplementary Propositions S12–S14 give the proof, exact-source rank reception and error propagation. Finite labels also require separately received force and Taylor-tail allowances. This rank calculation neither receives those allowances nor proves a target twist condition. The prior $W_* = \beta E_0$ fixes the additional jet directions through its global form and retains the source-family KAM conclusion below.

3.5 Training and precision-weighted coefficient fitting

3.5.1 Supervision of energy and dissipation The energy head is supervised by the eleven observed conservative force labels and their derived coefficient intervals. Its point prediction is the normalized gradient of $F_N(Vq)$, including the elastic-energy feature derivative in (10). The dissipation head uses the odd part of paired weak momentum residuals on the 22 full-coordinate records. Each signed record supplies four 64-panel windows. An individual weak residual includes momentum endpoints and an integral of conservative, dissipative and magnetic forces. The signed difference emphasizes the velocity-dependent term but retains the actual imperfect initial symmetry and numerical observations. It is not assumed to be an exact isolated damping measurement.

The conservative contribution is detached during dissipation updates. Thus energy and dissipation are fitted by specified block objectives; the whole procedure is not described as complete gradient descent on a single coupled selection score. All eleven configurations are development data. Neither their 257 temporal samples nor their velocity signs are counted as independent configuration samples.

3.5.2 Full-layer fitting and its high-order failure The network is initialized from a preceding single-configuration fit that performed 200 Adam updates and selected step 191. Eleven-configuration full-batch training then performs 2000 Adam updates at learning rate 10^{-3} and seed 43, selecting step 2000 by the combined same-data score. Both real and imaginary parts of all three weight layers in both branches change. The maximum normalized point-force residual is approximately 1.563110×10^{-5}. Direct outward evaluation nevertheless finds a quartic coefficient discrepancy larger than 6.345877×10^{-4} against the declared quadratic material family. This comparison is made after training; analytic zero cubic or quartic values are not substituted as labels.

Adding (34) reduces that high-order discrepancy. The energy branch uses one strong-Wolfe LBFGS iteration per outer cycle, and the dissipation branch one Adam step at fixed current energy. All 200 prescribed outer cycles are completed, with 432 energy closure evaluations and 200 dissipation updates. The coefficient loss falls from about 50743.76 to 4.399638. However, the point-force residual rises to approximately 2.434500×10^{-4}. In this stage the point residual is divided by the observed signal RMS whereas the coefficient residual is divided by its much smaller received error. A small coefficient objective therefore does not ensure an adequate finite-node fit under that scaling.

3.5.3 A linear block in the unchanged complex graph Fixing the two trained hidden layers makes the energy correction linear in c_E and the 24 real coordinates of its final complex weight. The centered output bias is a gauge and remains fixed. Let $\xi \in \mathbb{R}^{25}$ contain these coordinates

and $A_\psi \xi \in \mathbb{R}^{64}$ contain the 33 force and 31 coefficient predictions. The final calibration solves the numerical least-squares problem

$$\min_{\xi} \left\| D^{-1} \left(A_\psi \xi - \begin{pmatrix} \widehat{y} \\ \widetilde{c} \end{pmatrix} \right) \right\|_2^2, \qquad D = \operatorname{diag}(\eta, \widetilde{s}). \tag{39}$$

Here $\widetilde{c}$ and $\widetilde{s}$ are the saved coefficient centers and scales including center-rounding allowances. Each term is $(\text{residual}/\text{scale})^2$. There is no ridge penalty, and the force scales are their own received observation errors. The numerical design has rank 25 and condition estimate 2.84×10^6. These are floating-point diagnostics, not an exact-rank or exact-optimality proof. The exact-real fit is verified after saving the selected weights.

This is one final-readout solve and zero gradient updates. Only c_E and the final energy weight change; all other coordinates, including the entire dissipation head, remain identical to the preceding trained state. The model retains its 918-parameter graph. The method and scaling were chosen after observing the preceding fit and are reported as adaptive development, without a blind validation claim. The 64 rows in (39) do not represent 64 independent observations. Supplementary Section S10.1 proves the linear-block identity for the same centered network.

3.5.4 Independent reception of the selected parameters

Arb evaluation at 512 bits encloses the exact-real outputs defined by the saved binary weights, saved scales and exact native operators. At all eleven ideal rational nodes the 33 force enclosures satisfy their respective received intervals. All 31 coefficient enclosures also satisfy the original data intervals. The maximum normalized point-force residual is below 1.797797×10^{-8} and its largest ratio to its received allowance is below 0.790301. Including the admitted observation error gives the finite-node candidate-to-target force bound 5.023184×10^{-8}. This bound is restricted to the eleven configurations and to the declared observation/target assumptions.

The independently computed loss upper bound is $\overline{L}_{\text{jet}} < 0.01082009$. Table 2 distinguishes data-based guarantees from the tighter evaluation available only for the declared material formula. In particular, the smaller evaluator error must not be presented as the information content of the observed coefficient intervals.

The final energy skip is about 0.06002256805 and the norm of the 24 real readout coordinates is about 0.0017053353. The raw skip is not itself an identified material factor, because the neural energy-feature response also contributes to the quadratic term. An exact rational feasibility calculation admits the scalar comparison family $W(q) = \zeta q^T K_3 q/2$ with nonempty compatible interval approximately $[0.0599999981693, 0.0600000004402]$. An exact rational midpoint satisfies all 33 force and 31 coefficient constraints without assigning the material value to ζ. Thus the finite data do not require a nonlinear neural correction. Detailed source-bound coefficient and feasibility calculations appear in Supplementary Section S11.1.

Table 2. Upward bounds on normalized ordinary correction-coefficient error. The first numerical column uses the data intervals and the triangle inequality. The second additionally evaluates the declared quadratic simulation target.

Degree	Data-based bound	Declared-family evaluation
2	4.616863×10^{-7}	3.635946×10^{-8}
3	8.174276×10^{-7}	1.768735×10^{-8}
4	2.066906×10^{-6}	3.908709×10^{-8}

3.6 Controlled real–complex and reference-energy comparison

We compare four refitted models on the same full transverse mechanical system. The labels R and C denote real and complex networks with no fixed quadratic contribution to their output energy; GR and GC include that contribution. Every model uses the same kinetic energy, magnetic potential, magnetic input port, feature normalization, and positive-semidefinite dissipation construction. In particular, all four models receive the three mass-projected configuration coordinates and the normalized reference energy as inputs. The reference stiffness is consequently available as an input feature even when its fixed output contribution is omitted.

Writing $\chi = 0$ for R and C and $\chi = 1$ for GR and GC, the material energy is

$$\begin{aligned} U^m_{\chi,\vartheta}(x) = (\chi + c_E) E_0(x) \\ + E_{\text{net}}\big[N_\vartheta(s(x)) - N_\vartheta(0) \\ - D_{s_q} N_\vartheta(0) s_q(x)\big]. \end{aligned} \tag{40}$$

Here s_q consists of the first three components of s. The coefficient c_E is trainable in every cell. Thus the transformation $(\chi, c_E) = (1, c) \mapsto (0, c+1)$ preserves the represented energy exactly. The reference-energy contrasts measure the effect of residual initialization and the resulting optimization trajectory; they do not measure a difference in the admissible function classes.

The original complex implementation stores 918 real coordinates, including four output-bias coordinates that cancel in the centered energy and dissipation readouts. Removing these exact gauges leaves 914 trainable real

coordinates and preserves its represented functions. The comparison uses this gauge-free parameterization. Each complex head has two hidden layers of width 12. The real energy head has hidden widths 19 and 17, and the real dissipation head has widths 16 and 20. Their total parameter counts are equal: the complex energy and dissipation heads contain 457 coordinates each, while the real heads contain 453 and 461, respectively, including their scalar coefficients. Both real and imaginary coordinates of the complex weights are optimized; the complex activation is the same split hyperbolic tangent used by the full model.

Supervision is common to all cells. The total material-force labels are the observed correction-force labels plus the declared reference elastic force. Total Taylor-coefficient labels are obtained by adding the known reference quadratic polynomial to the received correction coefficients. The 31 coefficient rows are derived from the same 33 force components at 11 configurations; they are not independent observations. Dissipation uses the 22 signed full-field trajectories and the same four weak integration windows per trajectory. Energy contributions are held fixed when differentiating the dissipation loss.

Let r_j^F denote the total material-force residual and r_{awj}^D the odd signed-pair weak residual, both multiplied by ℓ/E_{net}, with $\ell = 10^{-3}$ m. The weak residual uses the three V-projected momentum channels, retaining the full-field configurations in every force and dissipation evaluation. For the reported dimensionless errors, define

$$\begin{aligned} J_F &= \frac{1}{33}\sum_{j=1}^{33}(r_j^F/s_F)^2, \\ J_c &= \frac{1}{31}\sum_{j=1}^{31}(r_j^c/s_j)^2, \\ J_D &= \frac{1}{132}\sum_{a=1}^{11}\sum_{w=1}^{4}\sum_{j=1}^{3}(r_{awj}^D/s_D)^2. \end{aligned} \tag{41}$$

Here r_j^c is the ordinary normalized coefficient residual and s_j is its received component scale. The shared force-label RMS is $s_F = 1.352136950992611$, and the shared nominal-damping RMS is $s_D = 0.004435135625933446$. These definitions govern the reported force, jet and weak MSE columns. The final precision-weighted readout instead divides each force component by its own received error η_j, as in (39); it does not use s_F for that solve.

For each of three predeclared seeds, all four cells are trained from fresh weights using the same 2000 point-force/weak-residual Adam updates, 200 point-force/Taylor-coefficient/weak-residual updates, one precision-weighted linear readout solve, and 200 final dissipation updates at fixed energy. Each cell receives 2200 energy and 2400 dissipation Adam backward steps. The readout design requires 64 additional output-component differentiations per cell. The linear solve updates the scalar elastic coefficient and the final energy-readout weight, with the hidden layers fixed and without ridge regularization. The known χE_0 contribution is retained as an affine offset. With total-material label vector y_{tot}, the solve is

$$\min_{\xi} \left\| D^{-1}(A_\psi \xi + b_\chi - y_{\mathrm{tot}}) \right\|_2^2, \tag{42}$$

where b_χ contains the known reference-force and quadratic-coefficient contributions. It has 18 unknowns for the real energy head and 25 for the complex head. We report results before and after this solve separately, together with its numerical rank and singular values.

For each error metric, the five within-seed contrasts are

$$\begin{gathered} R - GR, \quad C - GC, \quad R - C, \quad GR - GC, \\ (GR - GC) - (R - C). \end{gathered} \tag{43}$$

All three seeds are retained at every reporting stage. The existing 11 configurations are development data. The KAM certificate for the original 918-coordinate checkpoint does not transfer to the newly refitted comparison weights.

3.7 Empirical effects of representation, initialization and readout

All twelve predeclared runs complete the fixed budgets, with no discarded seed or repeated training attempt. Both branches and all six weight layers change; every complex weight layer changes in its real and imaginary coordinates. The linear blocks have numerical ranks 18 and 25 for the real and complex models, respectively. These ranks describe the numerical designs. The original certified 918-parameter checkpoint is unchanged.

Table 3 separates the optimization stages. The mean scaled jet errors before readout are approximately 8.57×10^4, 5.20×10^4, 25.65 and 73.53 for R, C, GR and GC. The fixed quadratic offset substantially changes finite-budget optimization, even though the free coefficient makes the two corresponding function classes equal. GR has smaller pre-readout jet error than GC in all three seeds. The joint coefficient objective also raises the R/C point-force MSE relative to the preceding point-only fit, from about 1.44×10^{-6} and 8.12×10^{-7} to about 1.39×10^{-3} and 1.44×10^{-3}, respectively. Its conditioning therefore affects finite-node accuracy as well as the jet.

The common final-readout rule reduces the jet means to approximately 0.01658, 0.01129, 0.01128 and 0.01039. In floating-point evaluation, every final model falls within all 33 force allowances and all 31 coefficient radii. An independent numerical implementation reconstructs the final forces, ordinary coefficients and all 88 weak windows for each model. These finite numerical checks have a different status from the outward certificate for the original checkpoint. The readout reduction is substantial in every cell; its different numbers of unknowns are part of the disclosed comparison.

Table 4 reports all five paired contrasts. The complex cells have smaller final jet means, but the unregularized readout and parameter allocation prevent attributing this difference to complex arithmetic alone. For final weak MSE, C has a larger mean than R, and their within-seed difference changes sign. GC has a smaller weak residual than GR in these runs. The evidence thus supports effects of initialization, representation and the final fitting block on this fixed development design, with no consistent ordering across stages and metrics. The three seeds measure optimization sensitivity and do not constitute independent physical samples.

3.8 Prediction at three new COMSOL configurations

The identities of all twelve comparison checkpoints were frozen before generating three new configurations with paired initial velocities. The prescribed configurations are $x = V\ell z$, with $\ell = 10^{-3}$ m and

$$\begin{aligned} z^{(1)} &= (1/3, 1/6, -1/3)^T, \\ z^{(2)} &= (-1/3, -1/6, 1/6)^T, \\ z^{(3)} &= (1/6, -1/3, 1/3)^T. \end{aligned} \tag{44}$$

Their prescribed velocities are $\dot{x} = \pm 25 V\ell z\,\mathrm{s}^{-1}$. The resulting six records cover $0 \le t \le 20\,\mu$s at 257 equally spaced output times. Each prediction starts from its record's observed full initial configuration and momentum, with $p = Mv$. The prescribed magnetic input is obtained by linear interpolation of the recorded field. No model is refitted, calibrated, or selected using these records.

We integrate the saved forced and damped model in the complete phase space $(x, p) \in \mathbb{R}^{64} \times \mathbb{R}^{64}$, using its learned conservative potential, its positive-semidefinite damping construction, and its known magnetic port. The implementation retains the derivative through the fourth, reference-energy input feature. For the mechanical reference norm

$$\|(u, \pi)\|_{\mathrm{m}} = \left(u^\top K_0 u + \pi^\top M^{-1} \pi\right)^{1/2}, \tag{45}$$

put $\Delta y_{kr} = (\widehat{x}_{kr} - x_r, \widehat{p}_{kr} - p_r)$. The primary score for checkpoint k and record r is

$$\eta_{kr} = \max_{0 \le j \le 256} \frac{\|\Delta y_{kr}(t_j)\|_{\mathrm{m}}}{\|(x_r(0), p_r(0))\|_{\mathrm{m}}}. \tag{46}$$

Each checkpoint score averages all six record scores with equal weights. Time RMS, final-time, and tip errors are retained with the individual trajectories.

The fixed primary DOP853 setting uses relative tolerance 10^{-10}, absolute tolerance 10^{-12}, and maximum step $0.5\,\mu$s. A second integration uses 10^{-12}, 10^{-14}, and $0.25\,\mu$s, respectively. We report the maximum normalized mechanical-norm difference between these predictions as a numerical resolution diagnostic. All 72 record predictions and their second-setting integrations completed; every configuration and velocity pair is retained.

For each observed record we also evaluate the complete 64 momentum equations without acceleration labels. In each of four consecutive 64-panel windows, normalized Simpson averaging gives

$$\begin{aligned} r_{krw} = \frac{M(v_r(t_b) - v_r(t_a))}{t_b - t_a} \\ + \mathcal{S}_w[\nabla U_k(x_r) + a_k(x_r) D_0 v_r \\ - \mu B_r \cos(b^\top x_r) b]\,. \end{aligned} \tag{47}$$

where U_k includes the known magnetic potential and $\mathcal{S}_w$ approximates a time average. We use the mass-dual norm $\|r\|_{M^{-1}} = (r^\top M^{-1} r)^{1/2}$ and normalize by $F_{0r} = \|K_0 x_r(0)\|_{M^{-1}}$. The full weak score averages these normalized norms over the four windows and six records. The paired-odd score first takes half the difference between the two signed residual vectors for each configuration, normalizes by the mean of their two F_{0r} values, and then averages the four windows and three configurations. These are norm scores, rather than the development-data mean squared losses.

The primary mean scores are 1.9510×10^{-7} for R, 1.9561×10^{-7} for C, 1.9227×10^{-7} for GR, and 1.9199×10^{-7} for GC. The corresponding two-setting differences are approximately 1.2×10^{-13}. The within-seed primary contrast R$-$C has mean -5.079×10^{-10} and sample standard deviation 2.69×10^{-10}; GR$-$GC has mean 2.787×10^{-10} and sample standard deviation 5.75×10^{-9}. Thus the two representation contrasts do not show a

Table 3. Four-cell development-data comparisons over all three fixed seeds. Entries are mean (sample standard deviation). Panel (a) separates the joint-training result from the final energy-readout calibration; panel (b) shows the additional dissipation updates at fixed calibrated energy. The real and complex readout blocks have 18 and 25 unknowns, respectively. These are floating-point residual diagnostics, not physical error certificates.

	Force MSE		Jet MSE	
Cell	Before readout	After readout	Before readout	After readout
R	$1.39\,(0.268)\times10^{-3}$	$1.71\,(0.0902)\times10^{-17}$	$8.57\,(2.92)\times10^{4}$	$1.66\,(0.02)\times10^{-2}$
C	$1.44\,(0.0628)\times10^{-3}$	$1.01\,(0.0653)\times10^{-17}$	$5.2\,(2.75)\times10^{4}$	$1.13\,(0.0218)\times10^{-2}$
GR	$5.98\,(3.07)\times10^{-9}$	$1.36\,(0.131)\times10^{-17}$	$2.57\,(1.15)\times10^{1}$	$1.13\,(0.0445)\times10^{-2}$
GC	$5.37\,(2.11)\times10^{-9}$	$1.03\,(0.0849)\times10^{-17}$	$7.35\,(6.68)\times10^{1}$	$1.04\,(0.0297)\times10^{-2}$

Cell	Weak MSE after readout	Weak MSE after dissipation updates
R	$1.76\,(0.537)\times10^{-5}$	$6.79\,(9.73)\times10^{-8}$
C	$1.64\,(0.431)\times10^{-5}$	$1.03\,(0.867)\times10^{-7}$
GR	$1.03\,(1.33)\times10^{-8}$	$8.52\,(10.8)\times10^{-9}$
GC	$3.79\,(0.793)\times10^{-10}$	$2.83\,(0.916)\times10^{-10}$

Table 4. Final within-seed contrasts: mean (sample standard deviation) over all three seeds. Each difference is formed before aggregation. Positive R–C or GR–GC favors the complex cell for that error metric. The reference-energy contrasts concern residual initialization within equal function classes.

Contrast	Force MSE	Jet MSE	Weak MSE
R-GR	$3.48\,(1.57)\times10^{-18}$	$5.3\,(0.246)\times10^{-3}$	$5.93\,(10.5)\times10^{-8}$
C-GC	$-1.76\,(11.9)\times10^{-19}$	$8.98\,(4.63)\times10^{-4}$	$1.03\,(0.868)\times10^{-7}$
R-C	$6.94\,(0.992)\times10^{-18}$	$5.29\,(0.335)\times10^{-3}$	$-3.52\,(16.1)\times10^{-8}$
GR-GC	$3.29\,(1.6)\times10^{-18}$	$8.86\,(1.84)\times10^{-4}$	$8.24\,(10.9)\times10^{-9}$
(GR-GC)-(R-C)	$-3.65\,(1.98)\times10^{-18}$	$-4.41\,(0.304)\times10^{-3}$	$4.35\,(17.1)\times10^{-8}$

consistent advantage. The trajectories provide fresh short-time agreement with this declared numerical benchmark; COMSOL observation error has no rigorous bound here, so differences of this magnitude do not establish physical accuracy differences.

The original 918-coordinate checkpoint remains the object of the KAM certificate. The newly fitted 914-coordinate comparison checkpoints are the objects of the numerical prediction experiment above.

The received lowest frequency interval of the saved conservative graph corresponds to a period between 0.5351 and 0.5355 s. The observation horizon covers less than $1/26750$ of that period. It does not resolve low-mode oscillation decay or longer-time control performance.

4 Conservative learning error and local KAM certification

4.1 What the coefficient bound can certify

For the restriction (5), known kinetic and reference magnetic terms are shared between the learned and target Hamiltonians. Their vector-field difference therefore depends only on the conservative correction force. The learning variable can be three-dimensional configuration, even though the canonical phase space has six real dimensions. This does not reduce the mechanical number of degrees of freedom. Under independent configuration sampling and suitable class and regularity conditions, the configuration-risk estimates in Supplementary Section S3 provide one connection with the nonzero-loss theory of Chen et al. (2022).

The deterministic bound (35) supplies a different, finite-data interface. Multiplication by $E_s\ell^{-|\alpha|}$ gives physical ordinary monomial coefficients; the derivative $D^\alpha W(0)$ additionally includes the factor $\alpha!$. The resulting Hessian, cubic and quartic intervals could support a physical-target KAM conclusion only if they imply the required spectral, nonresonance and twist conditions uniformly, together with the target's analytic hypotheses. Without additional constitutive structure, the present quartic intervals do not establish that implication. Their radius is much larger than the largest prescribed normalized magnetic quartic coefficient, approximately 5.363314×10^{-9}. All fifteen admitted total quartic coefficient intervals contain zero. This sign-ambiguity diagnostic is not itself a calculation of the Birkhoff twist.

4.2 A criterion for the three-coordinate restriction

The selected network can also be verified as a Hamiltonian in its own right. Suppose $U_N(Vq)$ has a realified holomorphic continuation near the origin, its Hessian is positive definite,

Table 5. Frozen-model prediction on three new COMSOL configurations with paired velocities over 0–20 μs. Entries are mean (sample standard deviation) over all three seeds; primary and full-weak scores average six records, while paired-odd scores average three configurations. The primary score is the maximum normalized full128 mechanical-norm error per record. The two-setting difference compares the fixed DOP853 predictions; it is a resolution diagnostic, not a certified solver-error bound. Full64 and paired-odd weak scores use the fixed mass-dual normalization.

Cell	Primary prediction	Two-setting difference	Full64 weak	Paired-odd weak
R	1.95×10^{-7} (5.71×10^{-10})	1.20×10^{-13} (3.30×10^{-15})	2.69×10^{-7} (2.00×10^{-9})	1.17×10^{-8} (1.44×10^{-8})
C	1.96×10^{-7} (6.94×10^{-10})	1.25×10^{-13} (1.68×10^{-14})	2.71×10^{-7} (1.81×10^{-9})	1.24×10^{-8} (4.17×10^{-9})
GR	1.92×10^{-7} (2.65×10^{-9})	1.17×10^{-13} (4.79×10^{-15})	2.59×10^{-7} (8.76×10^{-9})	4.53×10^{-9} (4.06×10^{-9})
GC	1.92×10^{-7} (3.31×10^{-9})	1.28×10^{-13} (1.16×10^{-14})	2.58×10^{-7} (1.09×10^{-8})	5.74×10^{-10} (1.95×10^{-10})

Table 6. Within-seed contrasts for the fresh short-time numerical observations, shown as mean (sample standard deviation) over three seeds. Every score retains all six records or all three signed configuration pairs. No model is refitted or selected using these observations.

Contrast	Primary prediction	Full64 weak	Paired-odd weak
R-GR	2.83×10^{-9} (3.21×10^{-9})	9.91×10^{-9} (1.07×10^{-8})	7.20×10^{-9} (1.81×10^{-8})
C-GC	3.62×10^{-9} (2.81×10^{-9})	1.25×10^{-8} (9.49×10^{-9})	1.18×10^{-8} (3.98×10^{-9})
R-C	-5.08×10^{-10} (2.69×10^{-10})	-1.62×10^{-9} (3.64×10^{-10})	-6.32×10^{-10} (1.83×10^{-8})
GR-GC	2.79×10^{-10} (5.75×10^{-9})	9.65×10^{-10} (1.89×10^{-8})	3.95×10^{-9} (3.88×10^{-9})
(GR-GC)-(R-C)	7.87×10^{-10} (5.70×10^{-9})	2.58×10^{-9} (1.89×10^{-8})	4.59×10^{-9} (2.19×10^{-8})

and its positive linear frequencies obey $k \cdot \omega \neq 0$ for integer vectors $0 < |k|_1 \leq 4$. The cubic terms are removed by the homological equation. With cubic generating function χ_3, the effective quartic term is

$$H_{4,\mathrm{eff}} = H_4 + \tfrac{1}{2}\{H_3, \chi_3\}. \tag{48}$$

Its action projection has the form

$$h_N(I) = \omega \cdot I + \tfrac{1}{2} I^T T_N I, \qquad \det T_N \neq 0. \tag{49}$$

Under these hypotheses, the classical local elliptic KAM argument gives a positive-measure family of invariant Lagrangian three-tori in every sufficiently small neighborhood of the equilibrium. Supplementary Proposition S20 gives the normal-form calculation and the scaling needed to apply Pöschel (2001). The finite frequency search is not an all-integer Diophantine assertion about the equilibrium frequencies: the Diophantine frequencies of the resulting tori vary in the nonlinear frequency image.

4.3 Same-checkpoint restriction result and its scope

The selected graph is pole-free on the explicit complex polydisc $|z_j| \leq 1/64$, with $q = \ell z$. Its positive mass and Hessian, finite degree-three/four divisors and cubic-corrected twist are received afresh from the same saved parameters. The frequencies are approximately

$$(11.73745102,\ 72.26615415,\ 326.09695871)\ \mathrm{rad\,s^{-1}}. \tag{50}$$

The twist has two positive and one negative eigenvalues and determinant near -5.271108×10^{19}, excluding zero. These checks meet the preceding qualitative criterion for this same conservative restriction. They do not specify a nonzero finite torus radius or certify a trajectory as an individual torus. The radius $1/64$ is an analytic-domain bound, not a torus radius.

This result concerns the learned conservative restriction with dissipation and time-dependent forcing removed. It establishes neither perturbative transfer to a physical target, invariance of the three-coordinate subspace inside the full rod, nor torus persistence under the dissipative controller. The following direct target-family calculation uses an additional constitutive hypothesis.

4.4 The neural Hamiltonian on all transverse coordinates

The full conservative candidate is defined by the same weights and exact source operators of Section 2.1:

$$H_N^{64}(x,p) = \tfrac{1}{2} p^T M^{-1} p + U_N(x), \qquad (x,p) \in \mathbb{R}^{128}. \tag{51}$$

This retains the full energy feature $E_0(x)$ and all 64 transverse configuration variables. It requires no invariance of the trial space. Let $P = \mathrm{diag}(1/Q)B$, $E = E_{\mathrm{net}}$, and write the realified energy readout as $N(l,e)$. Its actual Hessian is

$$\begin{aligned} U_N''(0) &= \alpha_N K_0 + E P^T N_{ll}(0) P + \kappa b b^T, \\ \alpha_N &= 1 + c_E + N_e(0). \end{aligned} \tag{52}$$

The energy-feature derivative is retained. Exact mass-metric bounds and spectral isolation prove 64 positive simple frequencies, with

$$|k \cdot \omega_N| \geq \frac{545883796425}{281474976710656} > 0.00193936\ \mathrm{rad\,s^{-1}} \tag{53}$$

for every integer k with $0 < |k|_1 \leq 4$.

The normal modes used for the nonlinear calculation are not accepted as exact floating-point eigenvectors. An exact proposal Gram matrix, Rayleigh residuals and separated spectral intervals give rigorous enclosures of the actual modal feature and tip projections. These enclosures propagate all energy-feature chain terms and the cubic homological correction into the full quartic action Hessian T_N^{64}. For $\Omega_N = \mathrm{diag}(\omega_{N,i})$, an outward product with a retained preconditioner B_0 proves

$$\left\| I - B_0 \Omega_N^{-1} T_N^{64} \Omega_N^{-1} \right\|_\infty < 2.353 \times 10^{-6} < 1. \tag{54}$$

Thus T_N^{64} is nonsingular. The realified graph is pole-free on a full complex configuration polydisc with displacement and rotation radii 2^{-40} m and 2^{-40} rad, respectively; momenta are entire. This is an analytic-domain certificate, not a torus-amplitude bound.

Supplementary Theorem S23 combines these compatible inputs. The saved graph has nonzero cubic terms, so its fourth-order normal form has a general fifth-order remainder. The small-action scaling applies classical KAM theory without an evenness assumption. Every open neighborhood of the equilibrium consequently contains a family of analytic invariant Lagrangian 64-tori whose union has positive 128-dimensional measure. The conclusion concerns this fixed conservative source graph. It includes no axial coordinates, damping, forcing, numerical torus radius, unknown-material transfer or superiority claim. Runtime operator rounding remains a separate implementation issue.

4.5 A direct local result for the admitted target family

Assume explicitly $\mathcal{W}_* = \beta E_0$, with β in the entire exact scalar compatibility interval of Section 3.5; its endpoints are approximately 0.0599999981693 and 0.0600000004402. The conservative three-coordinate target restriction is then

$$H_\beta^r(q, p_r) = \tfrac{1}{2} p_r^T A^{-1} p_r + \tfrac{1+\beta}{2} q^T K_3 q + \kappa[1 - \cos(d^T q)]. \tag{55}$$

Its cubic part vanishes. If u_j are mass-orthonormal linear modes, the quartic twist in physical canonical actions depends on $a_j = (d^T u_j)^2/\omega_j$ and obeys

$$\det T_\beta = -\frac{5\kappa^3}{512}(a_1 a_2 a_3)^2. \tag{56}$$

Uniform interval enclosures verify a positive simple spectrum, strictly positive modal port overlaps, and all frequency divisors through order four over the complete parameter interval. They give $|k \cdot \omega| > 11.73745099\,\mathrm{rad\,s^{-1}}$ for $0 < |k|_1 \le 4$, together with

$$-6.217103 \times 10^{17} < \det T_\beta < -6.217101 \times 10^{17}. \tag{57}$$

The twist has two positive and one negative eigenvalues throughout. Every admitted family member therefore has positive-measure families of invariant Lagrangian three-tori in sufficiently small neighborhoods of its equilibrium. Supplementary Corollary S25 supplies the uniform calculation and its analytic hypotheses.

This is a direct qualitative theorem for the declared family; it does not assert a numerical torus radius or an all-integer Diophantine property of the equilibrium frequencies. It also does not identify that family for an unknown device, prove perturbative transfer from the learned candidate, or establish invariance of the trial subspace within the full 64-coordinate system. The candidate certificate and the target-family certificate consequently answer distinct questions.

4.6 Source-conditioned extension to all transverse coordinates

The ideal benchmark's saved linear-elastic, small-rotation settings provide independent evidence for $\mathcal{W}_* = \beta E_0$ (Supplementary Section S2). For this constitutive family, retain all 64 transverse coordinates and define

$$H_\beta(x, p) = \tfrac{1}{2} p^T M^{-1} p + (1+\beta) E_0(x) + \kappa[1 - \cos(b^T x)]. \tag{58}$$

Over the entire original scalar compatibility interval, exact rational inertia calculations isolate all 64 positive linear frequencies. A degree-preserving finite-field coprimality certificate excludes zero modal tip overlaps. The uniform finite-divisor bound is

$$|k \cdot \omega(\beta)| \ge \frac{2472231}{1073741824} > 0.00230244\,\mathrm{rad\,s^{-1}}, \tag{59}$$

for every integer k with $0 < |k|_1 \le 4$. Writing $a_j = (b^T u_j)^2/\omega_j$ for mass-orthonormal modes, the quartic action Hessian is

$$T_\beta = \tfrac{\kappa}{8}\,\mathrm{diag}(a)(I_{64} - 2\mathbf{1}\mathbf{1}^T)\,\mathrm{diag}(a). \tag{60}$$

All a_j are nonzero. Hence T_β has inertia $(63, 1, 0)$ and nonzero determinant throughout the interval. Supplementary Proposition S26 gives the exact certificates. The small-action scaling in Supplementary Theorem S27 then applies Pöschel (2001) in dimension 64 and gives positive-measure families of invariant Lagrangian 64-tori arbitrarily near equilibrium.

These are tori in the 128-dimensional transverse phase space. The calculation uses neither invariance of the three-coordinate trial space nor a claim about 96-dimensional Lagrangian tori with axial dynamics. It assigns no numerical torus radius and makes no claim about damped or forced torus persistence, compiled-solver equivalence, or an unknown physical material. The independent full-coordinate neural certificate in Section 4.4 concerns a different Hamiltonian and uses its own cubic-corrected twist.

4.7 From the nonzero force-loss block to target-family KAM

The same prior allows the actual network fit to enter a sufficient target-family certificate. Let $g_{\theta,j}$ be its exact normalized force at observation component j, and use the received allowance η_j in the force block of the precision-weighted objective:

$$L_F(\theta) = \frac{1}{33}\sum_{j=1}^{33}\left|\frac{g_{\theta,j}-\widehat{y}_j}{\eta_j}\right|^2. \tag{61}$$

This is a force-block score on the existing development observations, distinct from the 31-coefficient loss. If $\overline{L}_F$ is the strict upper bound obtained from the saved output enclosures and the target force is βm_j, where m_j is the known observation multiplier for $j = 1, \dots, 33$, the triangle inequality gives

$$|g_{\theta,j} - \beta m_j| \le \eta_j\big(1 + \sqrt{33\overline{L}_F}\big). \tag{62}$$

For the same unchanged checkpoint,

$$0.03 < L_F \le \overline{L}_F < 0.037838 < 12/275. \tag{63}$$

Hence $11\eta_j/5$ is a valid allowance in (62). Intersecting its candidate-centered outer parameter strips gives an exact rational interval with outward decimal enclosure $[0.05999999283191808, 0.06000000327884982]$. The original data-only interval is contained in this larger interval; the learning score supplies a sufficient certificate without adding identification information to the observations.

The spectral calculation is repeated over the enlarged interval, giving $|k\cdot\omega| \ge 2646043/2147483648 > 0.00123215$ for $0 < |k|_1 \le 4$. The parameter-uniform certificate of nonzero modal overlaps and the twist formula (60) then give the same qualitative local 64-torus conclusion for every admitted target. Supplementary Propositions S29 and S30 prove the loss interface and its numerical instance.

This connects the actual nonzero force loss to a target-parameter set and then to a full-transverse conservative conclusion under the stated global constitutive prior. This loss-to-family construction does not itself prove the neural candidate's full-coordinate KAM property; the independent calculation in Section 4.4 supplies that result. The bound $12/275$ is a posteriori sufficient for this checkpoint and these observations; it is neither necessary nor a training stopping rule specified in advance. No generic nonlinear identification or complex representation advantage follows.

4.8 Coherent ambiguity and a nonzero material-jet neighborhood

The marginal sign ambiguity can be strengthened to a single coherent alternative. The material correction $\widetilde{W}(q) = \widetilde{\beta}E_0(q) + \kappa(d^Tq)^4/24$ admits one common parameter $\widetilde{\beta}$ satisfying all 33 original force-component and 31 derived coefficient intervals simultaneously. Its fifth material derivative is identically zero, while the retained magnetic cosine makes the total cubic and quartic Hamiltonian terms vanish. Its degree-four Birkhoff twist is therefore exactly zero (Supplementary Proposition S16). This coherent witness is admitted after allowing quartic material corrections; it does not contradict the explicitly quadratic family theorem. Independently, an even analytic correction with a nonzero higher-order tail leaves every retained signed-node force exactly unchanged and cancels the same quartic jet (Supplementary Proposition S17). Thus precise finite forces do not replace an independent target-tail condition. Neither construction proves absence of all invariant tori.

A positive sufficient result permits general cubic and quartic material perturbations while fixing the quadratic jet exactly. Let η_3, η_4 bound their ordinary coefficients in the normalization $q = \ell z$, W/E_s. Including the cubic homological correction, the full admitted family retains nonsingular twist whenever

$$\begin{aligned} b_4\eta_4 + b_3\eta_3^2 &< 1,\\ b_4 &< 1.123431\times 10^{11},\\ b_3 &< 2.663083\times 10^{12}. \end{aligned} \tag{64}$$

The rectangle $(\eta_3, \eta_4) = (2^{-22}, 2^{-38})$ gives a bound below 0.561; the common radius $\eta_3 = \eta_4 = 2^{-38}$ gives a bound below 0.409. Supplementary Lemma S31 establishes the criterion, and independent real-coordinate normal-form calculations verify the numerical instance. These are nonzero sufficient coefficient neighborhoods for a qualitative local theorem, with no numerical torus radius assigned.

The observed material coefficient boxes, compared directly with the same quadratic-family reference, give radii approximately $8.168713334 \times 10^{-7}$ and $2.067036334 \times 10^{-6}$ for degrees three and four. Their sufficient inverse-margin majorant exceeds 2.32218×10^{5} and does not meet (64). These are material-to-family quantities, distinct from a candidate-to-target error bound. In addition, the finite-data quadratic uncertainty and the unknown material's target-tail and analytic assumptions have not been received by this fixed-quadratic theorem. The current learning-to-KAM transfer to an unknown physical target therefore remains unproved. Failure of this sufficient bound is not a conclusion that the target lacks tori.

5 Full-coordinate power and conditional magnetic safety

5.1 Power identity and positive dissipation

Let $H_N(x,v) = v^T M v/2 + U_N(x)$ and let $y = g(x)^T v = \mu \cos(b^T x) b^T v$ be the full-state magnetic port. Equations (3) and (10) give

$$\dot{H}_N = -a_N(x) v^T D_0 v + uy. \tag{65}$$

This identity uses the exact constant mass and the actual full state, including the energy-feature chain rule. It holds for all network parameters and is independent of the restricted KAM result. It expresses structure preservation, not identification of the physical dissipative force. No learned momentum correction enters $\dot{x} = M^{-1}p$, so the candidate and plant ports agree if their mass, input map and full state agree.

For the selected weights, $c_E \simeq 0.06002256805$ and $c_C \simeq 0.07237264643$. The condition $1 + c_E > 0$, bounded real split-tanh output and a quadratic completion prove real coercivity of the candidate energy. Exact native factorizations give $D_0 > (6/25)M\,\mathrm{s}^{-1}$. On the full-state feature box $|s(x)|_\infty \le 2$, exact saved-weight bounds give

$$a_N(x) > 0.3977230, \qquad D_N(x) > 0.0954535 M\,\mathrm{s}^{-1}. \tag{66}$$

These are same-checkpoint lower bounds. The feature box is not automatically the full configuration energy domain used below. Supplementary Section S18.1 gives the other feature radii and exact conventions.

5.2 A potential-value safety theorem for the physical plant

Consider a physical system with the same mass and magnetic map, potential U_*, locally Lipschitz damping $D_*(x) \succeq 0$, and no additional unmodeled work. Suppose on the closure of a bounded configuration domain Ω,

$$|U_N - U_*| \le \varepsilon_U. \tag{67}$$

For actual continuous feedback $u = -\mathrm{sat}_U(ky)$, $U, k > 0$, assume

$$x(0) \in \Omega, \quad H_N(0) + \varepsilon_U \le c, \quad \inf_{\partial\Omega} U_N - \varepsilon_U > c. \tag{68}$$

Then the physical trajectory exists for all positive time, remains in Ω and obeys

$$H_*(t) \le H_*(0) \le c, \qquad H_N(t) \le c + \varepsilon_U. \tag{69}$$

Indeed, $\dot{H}_* = -v^T D_* v - y\,\mathrm{sat}_U(ky) \le 0$. The initial value estimate gives $H_*(0) \le c$. A first exit at the configuration boundary would require $H_* \ge U_* > c$, contradicting the energy inequality. Compactness and positive mass bound the velocity and allow continuation. This proof requires a uniform potential-value allowance; a finite-node force fit alone does not supply it. Supplementary Theorem S33 states the precise smoothness and domain hypotheses and gives the complete proof.

The physical and learned storage derivatives need not coincide. Writing $G = \nabla(U_N - U_*)$ gives, along the physical plant, $\dot{H}_N = -v^T D_* v + uy + v^T G$. Consequently a learned-energy sublevel is not automatically invariant even when the physical energy decreases. The conclusion (69) uses the potential-value comparison and the physical storage, rather than assuming away this discrepancy.

5.3 A same-checkpoint conditional guard

For a numerical instance, assume the physical family

$$U_*(x) = (1+\beta)E_0(x) + \kappa[1 - \cos(b^T x)], \tag{70}$$

where the constant β lies in the exact scalar compatibility interval of Section 3.5. This is an explicit constitutive-family and interval assumption. Simulation-derived compatibility is not evidence that an arbitrary physical rod necessarily belongs to that family.

On $\Omega_4 = \{x : E_0(x) < 4E_{\mathrm{net}}\}$, recomputed global derivative bounds and outward origin evaluation for the

selected energy head prove

$$\sup_{\overline{\Omega}_4} |U_N - U_*| < 0.003838998456 E_{\mathrm{net}}, \tag{71}$$

$$\inf_{\partial\Omega_4} U_N > 4.236161085119 E_{\mathrm{net}}. \tag{72}$$

Set $c = 2.05E_{\mathrm{net}}$. The boundary guard has positive normalized margin 2.182322086663. With $h_0 = 2.046161001544E_{\mathrm{net}}$, the initial set

$$\mathcal{I} = \{(x, v) : x \in \Omega_4, \quad H_N(x, v) \le h_0\} \tag{73}$$

contains the origin and an open neighborhood. Thus (69) follows conditionally for every initial state in $\mathcal{I}$ under the specified continuous full-state feedback. The global energy-head Hessian, compliance and scalar-interval calculation are supplied directly in Supplementary Section S18.2.

All 22 finest signed records at the eleven configurations belong to this same fixed initial set. Each calculation uses its own saved $t = 0$ full-coordinate position and velocity, interpreted as exact binary64 values, with no replacement by Vq or by the opposite sign. The strict common bounds are

$$\begin{aligned} &0.237377387997 < H_N(x_0, v_0)/E_{\mathrm{net}}, \\ &H_N(x_0, v_0)/E_{\mathrm{net}} < 0.972247611362. \end{aligned} \tag{74}$$

$$\frac{c - \varepsilon_U - H_N(x_0, v_0)}{E_{\mathrm{net}}} > 1.073913390182. \tag{75}$$

Supplementary Corollary S36 gives the separate sign-wise bounds and native coordinate reception. These are numerical initial states, with no certified error bound to a true physical initial state.

5.4 A sampled-work bound and strict energy ledger

For sampled control, assume additionally the physical damping form $D_*(x) = a(x)D_0$, with locally Lipschitz a and $0 \le a(x) \le 6/5$. This upper operator bound is a physical hypothesis separate from (66). Let u_k be the actual total transverse field held for duration h_k, with $|u_k| \le U$. It replaces the preparation cosine drive. On $H_* \le c$, the full-state equation and the energy cap give an output-rate bound $|\dot{y}| \le L_y$. For the present native matrices and $U = 10^{-4}$ T, outward evaluation proves

$$L_y < 49.953396682 \ \mathrm{J\,T^{-1}s^{-2}}. \tag{76}$$

At a nonzero hold's actual start t_k, require a valid current observation $|y(t_k) - \widehat{y}_k| \le \varepsilon_k$. Since $\dot{H}_* = -a(x)v^T D_0 v + u_k y \le u_k y$, every hold prefix $0 \le s \le h_k$ below the energy cap satisfies

$$\begin{aligned} H_*(t_k + s) - H_*(t_k) &\le F_k(s), \\ F_k(s) &:= s(u_k\widehat{y}_k + |u_k|\varepsilon_k) \\ &\quad + \tfrac{1}{2}s^2|u_k|L_y. \end{aligned} \tag{77}$$

The omitted dissipation is nonnegative. This choice takes zero dissipation credit; it does not assume a zero derivative of the physical damping coefficient.

Initialize a trusted ledger by rounding an upper enclosure of $H_N(x_0, v_0) + \varepsilon_U$ upward to a fixed energy grid g, so $H_*(t_0) \le L_0 < c$. Write $\Delta_k = F_k(h_k)$. Accept a nonzero proposal only if its observation and actual field limit are valid and

$$\Delta_k \le \alpha_{\mathrm{res}}(c - L_k), \qquad 0 < \alpha_{\mathrm{res}} < 1, \tag{78}$$

$$L_{k+1} = \lceil \max\{0, L_k + \Delta_k\} \rceil_g < c. \tag{79}$$

Otherwise apply exactly zero and retain the ledger. An exactly zero total-field hold also uses this dissipative branch directly and requires no port-observation error bound. Starting with $H_*(t_k) \le L_k < c$, convexity of F_k yields $F_k(s) \le \max\{0, \Delta_k\}$. A hypothetical first time with $H_* = c$ would therefore satisfy $H_* \le L_k + \max\{0, \Delta_k\} < c$, a contradiction; the rate bound is used only before that first exit. At the hold endpoint, $H_*(t_{k+1}) \le L_{k+1} < c$. Induction gives the strict energy cap for any sequence of holds without finite-time accumulation, provided the observation, timing and total-field execution conditions hold at every step. Also $c/(1 + \beta_-) < 4E_{\mathrm{net}}$, so this physical cap keeps the configuration in Ω_4. Supplementary Section S19 gives the rate majorant, continuation argument and complete execution assumptions.

For a received initial-proposal calculation, fix $\varepsilon_y = 10^{-8}\,\mathrm{J\,T^{-1}s^{-1}}$, $g = 10^{-9}E_{\mathrm{net}}$ and $\alpha_{\mathrm{res}} = 1/2$. At each of the 22 saved numerical states, an outward port enclosure supplies a rational midpoint $\widehat{y}_0$ whose arithmetic error is below this allowance. The nonzero opposite-sign proposal $u_0 = -U\,\mathrm{sign}(\widehat{y}_0)$ passes (78)–(79) for all 22 states at both $1\,\mu$s and $10\,\mu$s. At $100\,\mu$s, all 22 proposals are rejected and the exact zero fallback retains their initial ledgers. The same cap, field limit, error allowance and rounding grid are used throughout; Supplementary Section S19.1 gives the corresponding ledger and work bounds.

The port-error allowance is conditional for a physical sensor, and this initial-proposal calculation alone executes no feedback rollout; Section 5.5 gives a separate numerical execution. Accepted holds may spend energy reserve; neither monotone energy decrease nor improved control

performance is asserted. The arbitrary-horizon theorem depends on valid sensing at each actual hold start and execution of each accepted total field with the specified timing. Physical sensing, actuator and timing guarantees remain to be supplied for an experiment.

5.5 Executed feedback and a continuous reference-IVP certificate

A new native COMSOL execution uses the selected network's potential guard to initialize the energy ledger and the common mechanical port to choose the field sign. Starting from a fresh $20\,\mu$s preparation solve, two approximately $10\,\mu$s feedback holds apply opposite nonzero total fields, followed by a scheduled zero-field hold. The next sign is computed from the current saved state; the preparation cosine is replaced during each hold. The executed magnitudes are the binary64 number immediately below 10^{-4} T. All 101 native solution and velocity entries, including the clock variables, agree bitwise at the three continuation handoffs. This is a numerical feedback execution, with no physical actuator or sensor involved.

For a continuous statement, define a reference full-coordinate IVP by $M = M_0 + M_r$, $K = (53/50)K_0$ and $D = \alpha_{\mathrm{shift}} M_0$, using the exact saved coefficients. It starts at the fresh seed's own saved numerical position and velocity and uses the same three executed fields and exact saved switching times. A piecewise cubic Hermite reconstruction and a residual estimate in the mechanical energy norm enclose this IVP without resetting the error at a switch. The reconstruction retains every control endpoint; each retained cell is bounded over its whole duration. Supplementary Section S19.2 gives the residual argument, and Section S19.3 specifies its numerical reception and the reconstruction choices, including unsuccessful bounds.

The preassigned port allowance remains $\varepsilon_y = 10^{-8}\,\mathrm{J\,T^{-1}s^{-1}}$. At the first two hold starts, outward bounds on the discrepancy between the reference-IVP port and the actual logged readout are respectively at most 10^{-50} and less than $6.874 \times 10^{-9}\,\mathrm{J\,T^{-1}s^{-1}}$. Both nonzero proposals therefore satisfy the required observation premise. The bound at the third start is approximately 1.640×10^{-8} in the same units and exceeds the allowance. That hold applies exactly zero total field, so $\dot{H}_* = -v^T D v \le 0$ and its energy argument needs no port observation. No three-observation accuracy claim is made.

For the same checkpoint, exact initial-ledger and command calculations give

$$\begin{aligned} L_0/E_{\mathrm{net}} &= 0.836050803, \\ L_1/E_{\mathrm{net}} &= 0.870553256, \\ L_2/E_{\mathrm{net}} = L_3/E_{\mathrm{net}} &= 0.910436431. \end{aligned} \tag{80}$$

Combining the two nonzero prefix-work bounds with the zero-field dissipation identity proves

$$H_*(t) \le 0.910436431 E_{\mathrm{net}} < 2.05 E_{\mathrm{net}}, \quad t_0 \le t \le t_3, \tag{81}$$

for this specified reference IVP. The proof uses the stated quadratic-family assumption, positive semidefinite damping, the same network's potential-value allowance, and the actual logged times and commands. It is not an enclosure theorem for COMSOL's compiled equations or dense output. Figure 2 displays the separate numerical execution. The ledger-based certificate for this finite execution supplies no physical sensor guarantee, arbitrary future feedback guarantee, or comparative control-performance claim.

5.5.1 An offline signed-work bound for the same execution

The received continuous comparison also bounds the tip angle $\theta = b^T x$ throughout every retained cell. For each actual constant total field u_k, the magnetic work has the exact primitive

$$\begin{aligned} H_*(t) - H_*(t_k) = &- \int_{t_k}^{t} v^T D v \, ds \\ &+ \mu u_k [\sin\theta(t) - \sin\theta(t_k)]. \end{aligned} \tag{82}$$

Exact Hermite ranges, the already received full-coordinate error radii and rational sine remainders bound this expression on all 192 retained cells. Omitting the nonnegative dissipated work gives the stronger offline certificate

$$H_*(t) < 0.832211815 E_{\mathrm{net}}, \qquad t_0 \le t \le t_3. \tag{83}$$

The three endpoint bounds are 0.829688104, 0.827405859 and 0.827405859 times E_{net}. On each hold the energy stays below its actual starting energy. The received port-sign bounds additionally prove energy is nonincreasing throughout the second and third holds; fifteen cells of the first hold fail that sufficient sign test. The smaller cap strengthens the certificate for this same finite reference history. The prospective controller, ledger and failed third sensor reception remain as stated above. Supplementary Section S20 gives the signed-work proof and its numerical reception. No new trajectory is integrated.

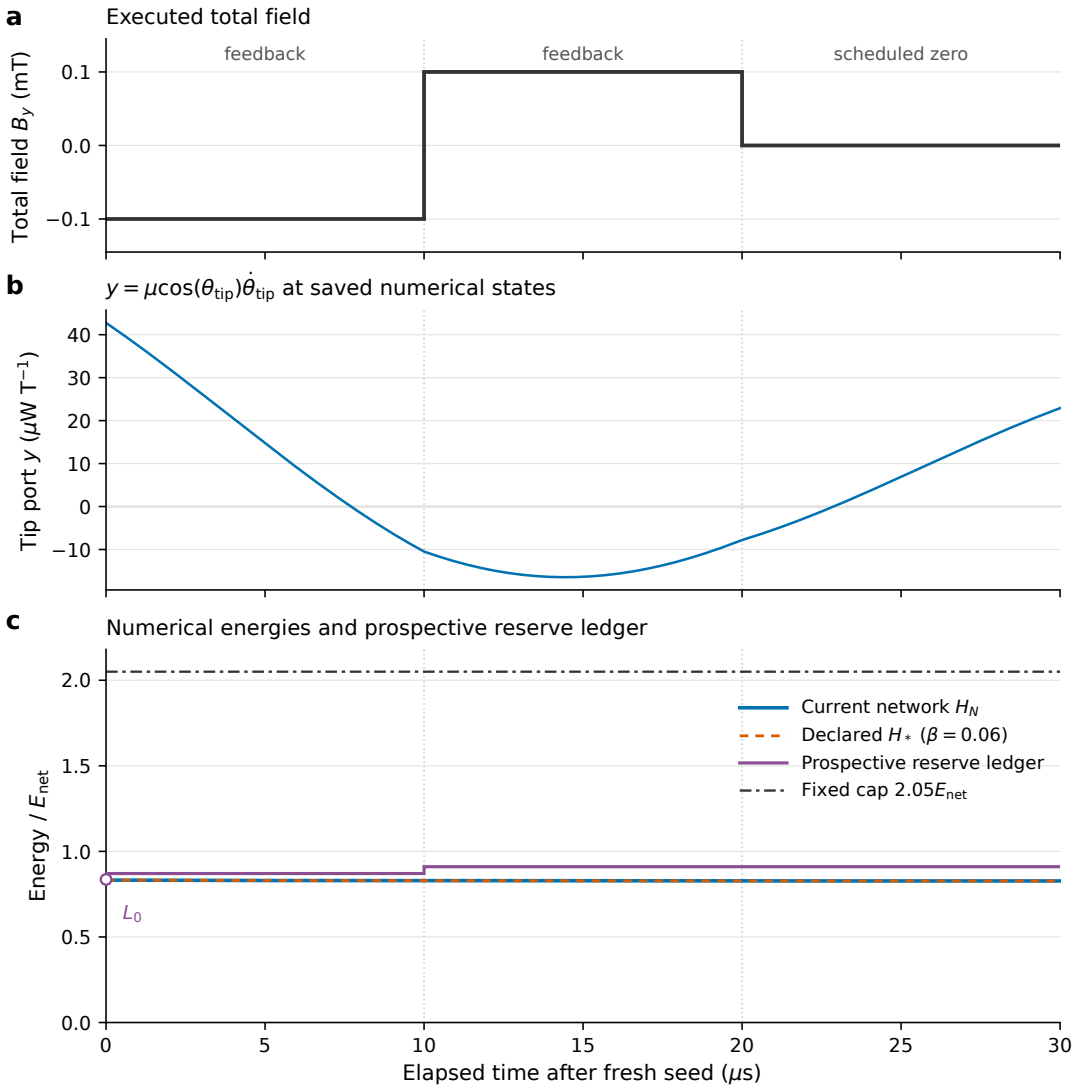


Figure 2. Executed total field, saved-state mechanical port, and numerical energies over the three approximately $10\,\mu$s holds, measured from the fresh numerical seed. The two nonzero fields are chosen from current observations; the third field is scheduled to zero. The learned and declared quadratic-family energies nearly overlap on this scale. The open marker denotes L_0; on each hold the reserve line displays $\max(L_k, L_{k+1})$. The curves connect saved numerical samples. The continuous cap in (81) is proved for the separately specified reference IVP by residual bounds and the work ledger, rather than inferred from these plotted samples.

6 Conclusions

This study connects finite observations, nonzero learning error and structural guarantees for a specified magnetic-rod model. Eleven paired configuration preparations supply 33 force components and 31 derived coefficient constraints. Full-layer training followed by a single precision-weighted readout calibration produces a saved complex network satisfying every received interval. Independent outward evaluation bounds its actual nonzero coefficient loss. The derived coefficient rows reuse the force data, so their joint objective represents deterministic conditioning rather than additional independent samples. An exact-source observability calculation gives ranks 31, 41 and 42 for the restricted labels, complete forces at the old nodes, and an augmented off-trial-space design, respectively. It specifies how full-field data reuse could reduce the additional measurements needed within the four-feature target class.

The same full-coordinate energy admits dimensionally consistent complex canonical coordinates and an explicit Störmer–Verlet difference map. Exact kinetic and potential subflows give symplecticity and self-adjointness. The domain-dependent error analysis proves fixed-time second-order convergence and separates integration error from uniform force-learning error and numerical implementation error. The received full spectrum also supplies a strict physical step condition for the equilibrium linearization. Full-coordinate finite-time calculations with the saved checkpoint exhibit the expected second-order convergence under the fixed refinement protocol.

For the unchanged saved weights and exact source operators, a positive equilibrium Hessian, finite nonresonance,

analyticity and the complete cubic-corrected twist establish local KAM for all 64 transverse coordinates. Every neighborhood of equilibrium contains a family of analytic invariant Lagrangian 64-tori whose union has positive 128-dimensional measure. This result requires no invariant trial space and assigns no numerical torus radius. Separately, under the declared quadratic constitutive prior, the actual nonzero force-loss budget $L_F \leq 12/275$ supplies a sufficient target-parameter interval on which the full-coordinate KAM conditions are verified. The candidate theorem and the prior-dependent target theorem concern different Hamiltonians.

Dissipation and magnetic actuation are treated by power rather than conservative KAM. The squared dissipation readout enforces nonnegative loss, while a potential-value allowance transfers a conditional energy guard to a plant with the specified mass and input port. All 22 saved numerical initial states satisfy the instantiated guard. A sampled-work bound charges held-input work to a strict energy ledger. A native three-hold numerical execution and its separately defined continuous reference IVP demonstrate the two nonzero-input and zero-input branches under the received observation and timing conditions. Signed-work reception tightens that particular reference-history cap to $0.832211815E_{\mathrm{net}}$ and proves energy is nonincreasing throughout the second and third holds. Physical use requires these conditions at each actual hold, including valid sensing errors and execution of the accepted total field.

The present observations also admit a scalar stiffness correction and leave the small quartic material signal unresolved. They consequently establish neither a need for the nonlinear network nor a complex-network advantage. Parameter-matched refits recover all five prescribed comparisons and show a substantial effect of the residual initialization and final readout, with no consistent real–complex ordering across stages and metrics. Six new COMSOL records at three configurations test the frozen models over $20\,\mu$s; their normalized full-state prediction scores are approximately 1.92×10^{-7} to 1.96×10^{-7}. These are short-time numerical benchmark comparisons, with no consistent complex advantage. The finite-data counterexamples and material-jet tolerance in Section 4 specify why candidate regularity and small node loss alone do not identify an unknown target's KAM conditions. Extending the result to an unknown device requires observations that resolve the relevant coefficients, independently justified constitutive remainders, and bounds on sensing, actuation and unmodelled work. These requirements connect the mathematical certificates to the measurements needed for a physical robotic test.

Data and code availability

The accompanying reproducibility manifest identifies the native simulation records, selected network parameters, exact operator records and interval verification sources used for each numerical statement. It separates local source files from workspace evidence and records their SHA-256 identities. The supplementary material supplies the mathematical proofs and arithmetic conventions. The eleven development configurations and six new off-design simulation records are indexed separately. The latter are generated and evaluated after all comparison weights are frozen. These simulation records are not physical measurements or independent population samples.

Declarations

This study reports mathematical analysis and simulated mechanical data. It involved no human participants or animals.

References

Abbott JJ, Diller E and Petruska AJ (2020) Magnetic methods in robotics. *Annual Review of Control, Robotics, and Autonomous Systems* 3: 57–90. doi: 10.1146/annurev-control-081219-082713

Chen Y, Matsubara T and Yaguchi T (2022) KAM theory meets statistical learning theory: Hamiltonian neural networks with non-zero training loss. *Proceedings of the AAAI Conference on Artificial Intelligence* 36(6): 6322–6332. doi: 10.1609/aaai.v36i6.20582

Desai SA, Mattheakis M, Sondak D, Protopapas P and Roberts SJ (2021) Port-Hamiltonian neural networks for learning explicit time-dependent dynamical systems. *Physical Review E* 104: 034312. doi: 10.1103/PhysRevE.104.034312

Greydanus S, Dzamba M and Yosinski J (2019) Hamiltonian neural networks. In: *Advances in Neural Information Processing Systems*, vol. 32. https://arxiv.org/abs/1906.01563.

Hairer E (2010) Geometric numerical integration. Lecture 2: Symplectic integrators. Lecture notes, TU München, January–February 2010. https://www.unige.ch/~hairer/poly_geoint/week2.pdf.

Haro À and Luque A (2019) A-posteriori KAM theory with optimal estimates for partially integrable systems. *Journal of Differential Equations* 266(2–3): 1605–1674. doi: 10.1016/j.jde.2018.08.003 Preprint version used: https://arxiv.org/abs/1711.05515v1.

Kim Y, Parada GA, Liu S and Zhao X (2019) Ferromagnetic soft continuum robots. *Science Robotics* 4(33): eaax7329. doi: 10.1126/scirobotics.aax7329

Pöschel J (2001) A lecture on the classical KAM theorem. In: *Proceedings of Symposia in Pure Mathematics*, vol. 69, pp. 707–732. doi: 10.1090/pspum/069/1858551 Preprint version used: https://arxiv.org/abs/0908.2234v1 (Version 1.3, 2009).

Trabelsi C, Bilaniuk O, Zhang Y et al. (2018) Deep complex networks. In: *International Conference on Learning Representations*. https://arxiv.org/abs/1705.09792.

Zhang SQ, Gao W and Zhou ZH (2022) Towards understanding theoretical advantages of complex-reaction networks. *Neural Networks* 151: 80–93. doi: 10.1016/j.neunet.2022.03.024

Zhong YD, Dey B and Chakraborty A (2020) Dissipative SymODEN: Encoding Hamiltonian dynamics with dissipation and control into deep learning. In: *ICLR Workshop on Integration of Deep Neural Models and Differential Equations*. https://arxiv.org/abs/2002.08860.